

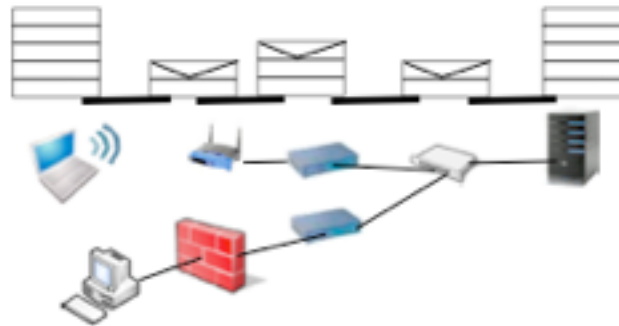
UCL

Université
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Computer Networking

Principles
Protocols
and
Practice



IP Multicast

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<http://inl.info.ucl.ac.be/>



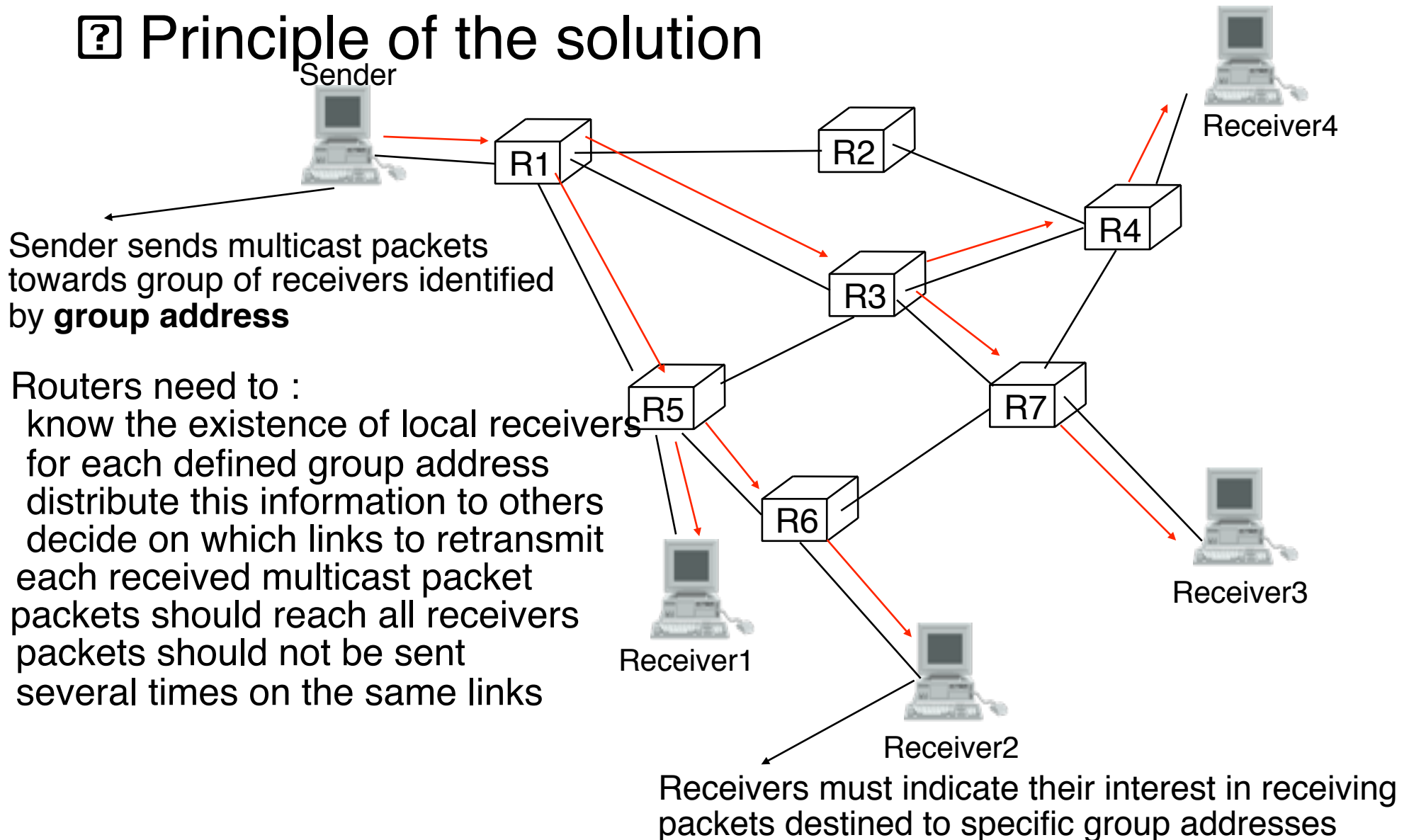
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The motivation for non-unicast

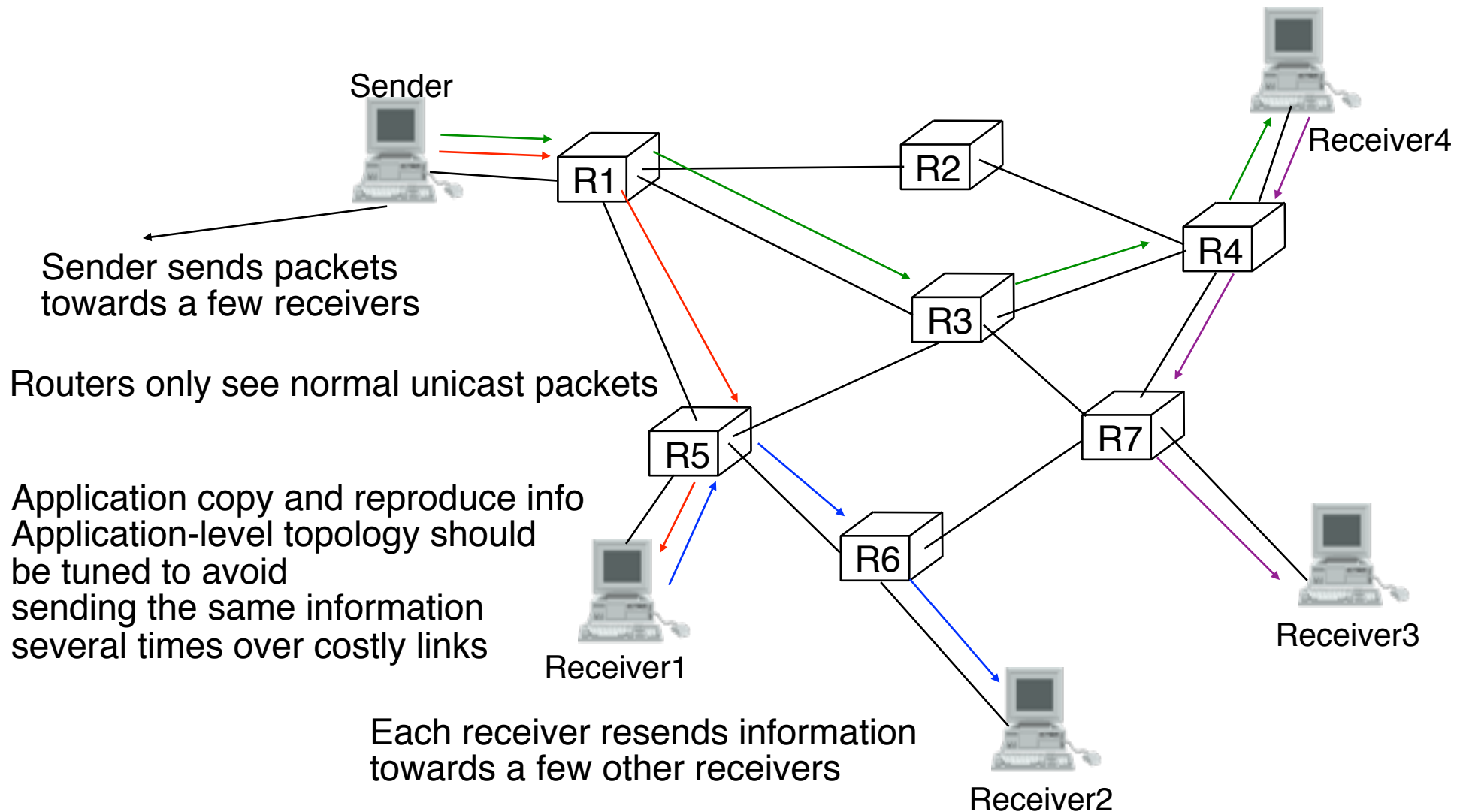
- Principles of unicast transmission
 - Source is identified by a unique address
 - Destination is identified by a unique address
 - Packets are sent from Source towards Destination
- Motivation for a different mode than unicast
 - How to transmit same info to set of receivers ?
 - Source-based solution
 - sender sends **N copies** of this information, one for each receiver
 - easy to implement, but consumes a lot of bandwidth
 - Network-based solution
 - sender sends **a single copy** of the information
 - **network (routers) takes care** of distributing this information efficiently towards all interested receivers
 - more efficient, but requires specific protocols and mechanisms

Network-level multicast

❓ Principle of the solution



Application-level multicast



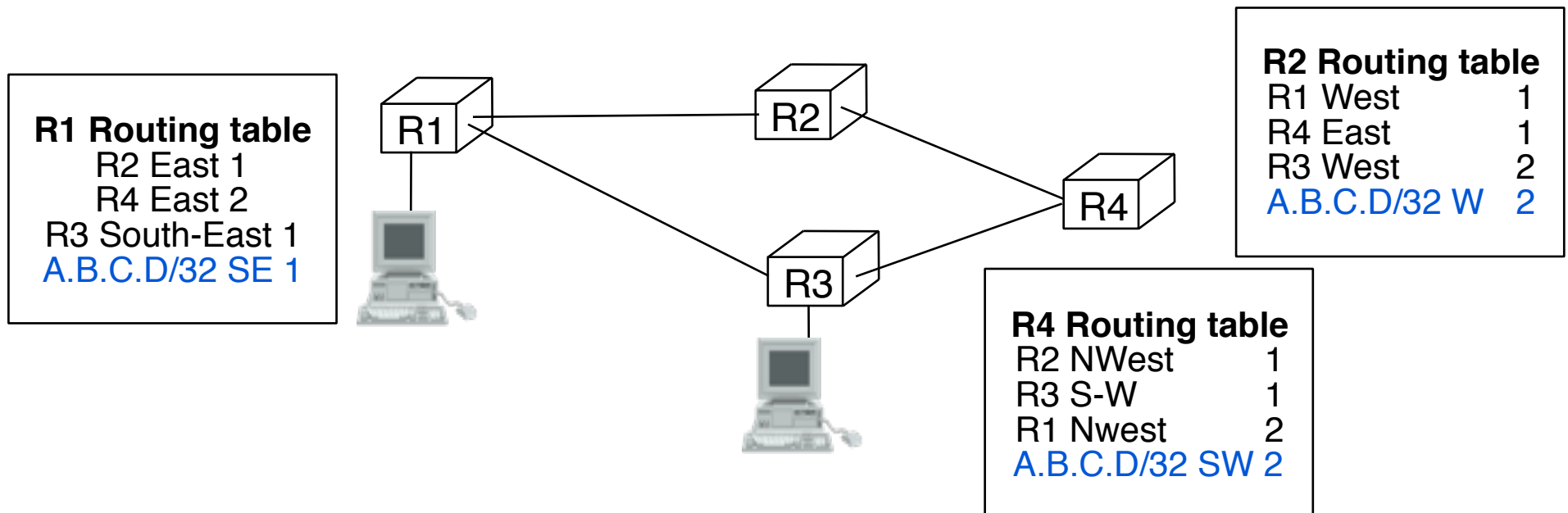
- Examples : Usenet news, IRC, some P2P apps, ...

Sample non-unicast applications

- Localization applications
 - a server announces its availability
 - a client tries to locate the closest server
- Distribution of real-time information
 - Multiparty conferences
 - Distributed games
 - Real-time Monitoring
 - Temperature
 - stock prices
- Reliable distribution of information
 - software updates
 - news information

Localization of services

- How to find the closest server for service Y ?
 - **Network-based anycast** service
- Network-level solution
 - IP address **A.B.C.D** reserved for service Y
 - each server uses **A.B.C.D** as logical address
 - routers distribute route towards **A.B.C.D/32**



Localization of services (2)

- How to find the closest server for service Y ?
 - **Name-based anycast** service
- Application-level solution
 - Name `serviceY.domain` used for service Y
 - Configure the DNS servers to answer requests for `serviceY.domain` with closest IP address
 - DNS server must have some information about network topology
 - DNS server may also take server load into account
 - Often used by content distribution networks (CDNs)

Distribution of realtime information

- Example : N-way videoconference with RTP
- Network-level solution
 - One multicast group for audio
 - One multicast group for video
 - One multicast group for shared whiteboard
 - All participants send data to everybody

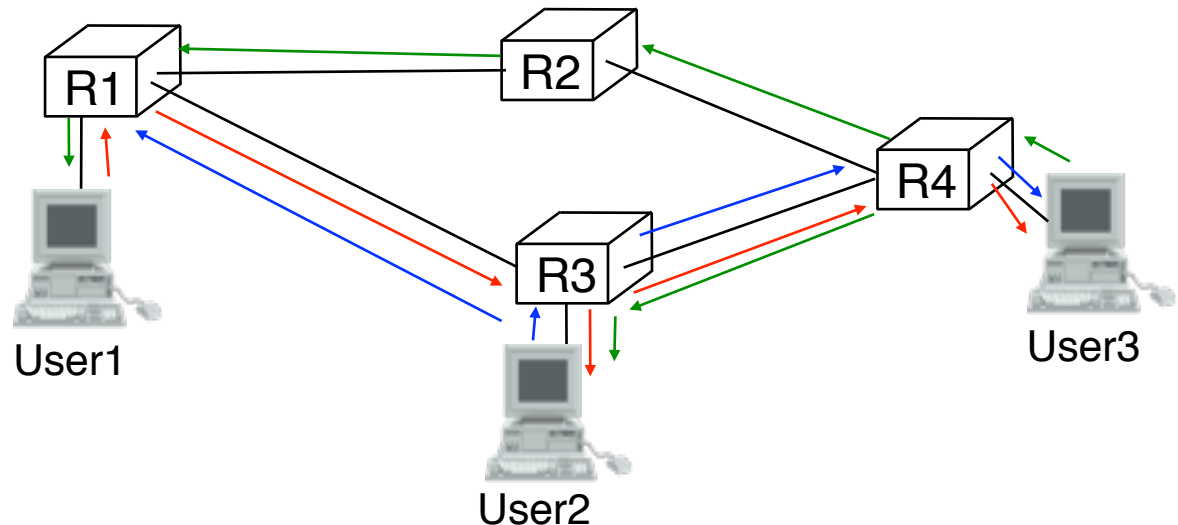
Number of senders : a few
Number of receivers : a few
Data traffic

continuous for video
may be bursty for audio
bursty for whiteboard

Lifetime : long

Volatility : stable

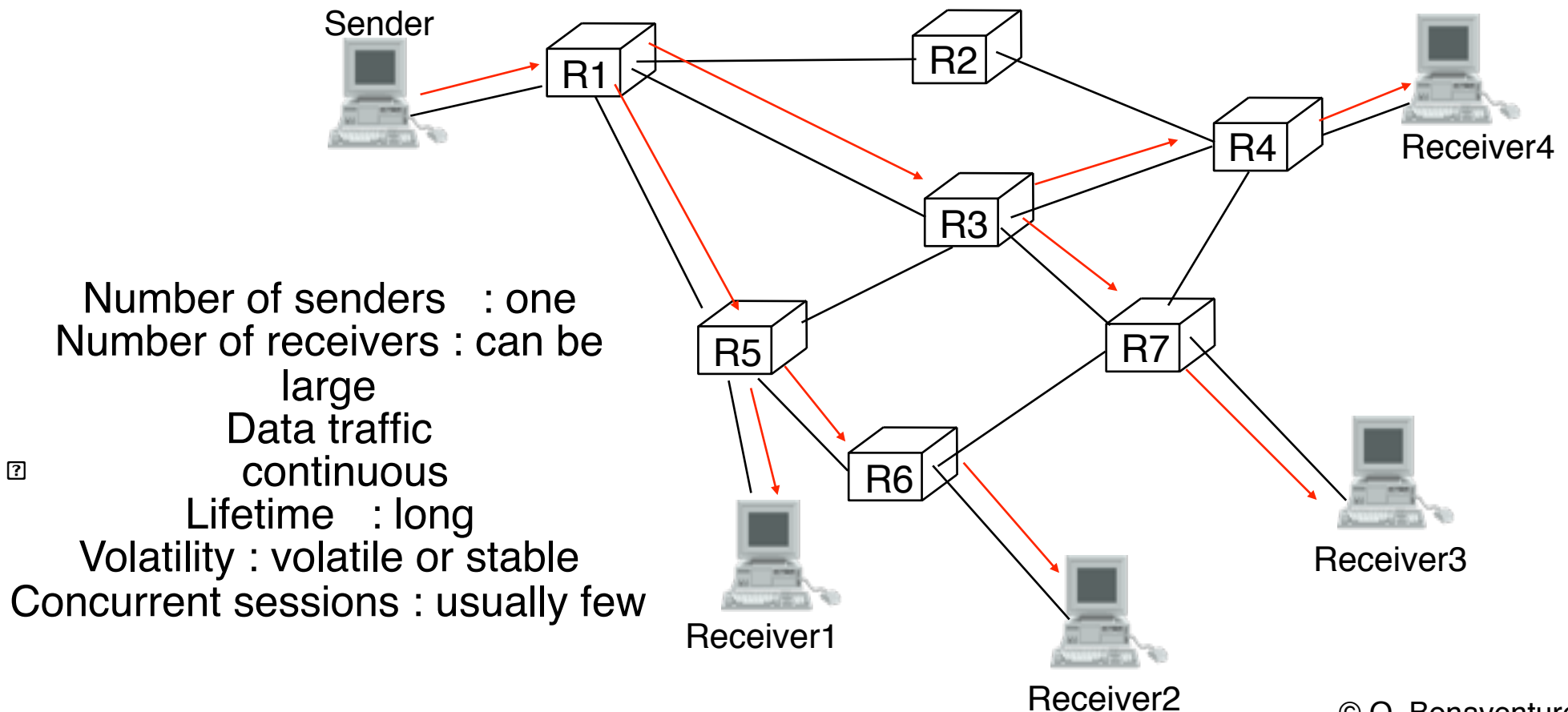
Concurrent sessions : can be large



A distributed game would be different

Distribution of realtime information (2)

- Live audio/video
 - one sender transmits streaming audio/video to large number of receivers



The benefits and costs of network-level multicast

- Benefits
 - Multicast allows a more efficient utilization of the available bandwidth
 - no unnecessary packet duplication
- But, to support IP multicast, routers must
 - implement special multicast routing protocols
 - implementation and configuration cost
 - maintain some state information for multicast
 - memory consumption
 - IP multicast packets require special treatment
 - CPU consumption

————→ The bandwidth saved should be compared to the router costs!

The Any-Source IP Multicast (ASM) service model

- Principles
 - A fraction of the IP address space is reserved exclusively for IP multicast
 - 224.0.0.0 -> 239.255.255.255
 - A **multicast address** identifies **a group of receivers**
 - receivers can join and leave a group at any time
 - members of a group can be located anywhere
 - **Any** source host can transmit IP packets towards **any** IP multicast address
 - a sender does not need to be a member of a group
 - Routers take care of the efficient transmission of multicast packets towards their destinations
 - Packet forwarding based on destination address

IP Multicast addresses

- Initial Allocation of IP addresses
 - Well-known addresses
 - 224.0.0.1 : all hosts
 - 224.0.0.2 : all multicast routers
 - 224.0.0.5 : all OSPF routers
 - Other addresses like 224.0.1.11
 - Reserved addresses to broadcast IETF meetings
 - Dynamic multicast address assignment
 - `sdr` application listens to special multicast group
 - this group is used to broadcast announcements of the creation of new IP multicast groups
 - When an application needs to create a multicast group, it requests `sdr` to select one address at random
 - `sdr` checks that it did not heard anything about this IP address
 - Suitable for research experiments, ...
more difficult in a commercial Internet

IP Multicast addresses (2)

- Multicast Address allocation
 - Evolution is to allocate addresses hierarchically
 - Administratively scoped addresses
 - 239.0.0.0 - 239.255.255.255
 - reserved for utilization inside private domains
 - Should not appear on the global Internet
 - Static group address assignment (GLOP)
 - 233.0.0.0 - 233.255.255.255
 - each Autonomous System receives 256 addresses in this range and can allocate those addresses to its users
 - MASC : Multicast Address Set-Claim
 - Dynamic Method to allocate blocks of IP multicast addresses to ISPs under development
 - Multicast Addresses allocated for some time only
 - far from being deployed

Evaluation of the ASM model

- Advantages
 - Any source can send packets to any group
 - Any receiver may join any group
- Drawbacks
 - How do we allocate IP multicast addresses ?
 - Random allocation is not really a long term solution
 - Risk of collision is severe
 - Lack of access control
 - Any source can send packets to any group
 - Consider a videoconference between professors where students could easily send packets to interrupt the conference...
 - Any receiver may join any group
 - Encryption can be used to protect packet contents, but ...
 - Inefficient handling of well-known sources
 - CNN uses 226.0.0.10 and IraqiTV selects same address...

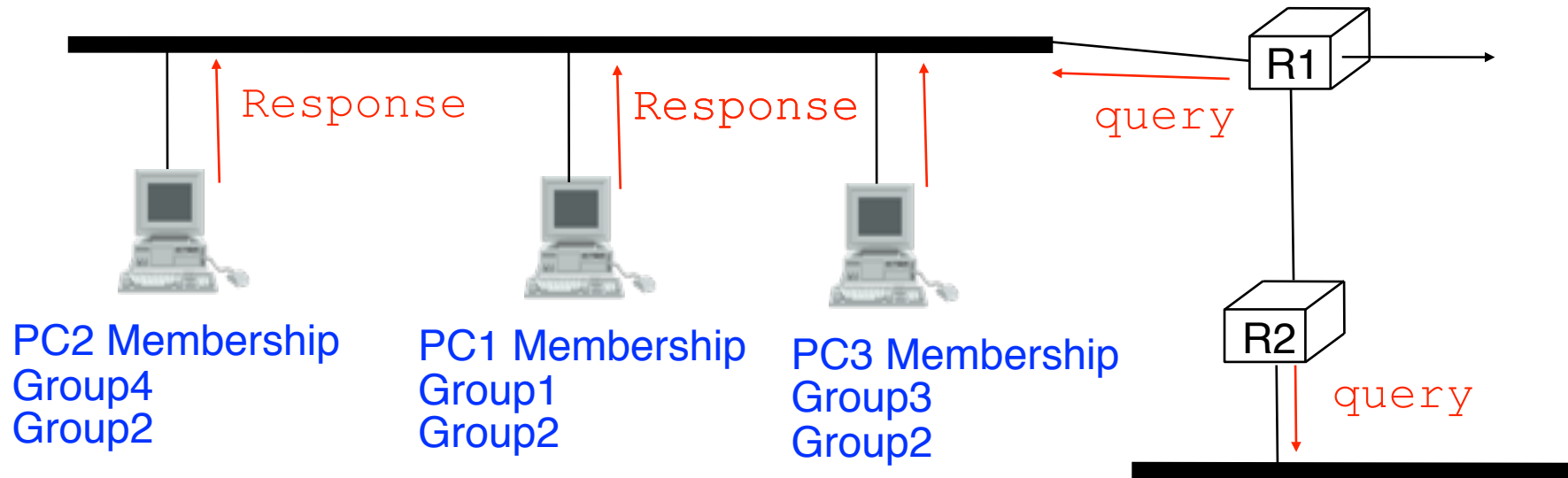
The Source-Specific IP Multicast (SSM) service model

- Objective
 - Provide a deployable **one-to-many** service
- Principles
 - **SSM multicast address = channel from a sender**
 - A single source can use as many channels as there are SSM addresses
 - Several senders can use same channel id without problems
 - A fraction of the IP address space is reserved exclusively for SSM multicast addresses
 - 232.0.0.0 -> 232.255.255.255
 - Receivers subscribe to channels from senders
 - Receiver knows from which sender packets will arrive
 - Routers retransmit multicast packets from the sender to all channel subscribers
 - Packet forwarding based on both packet source and destination addresses (Channel)

Issues for network-level multicast

- 1. How to localize all the members of each multicast group ?
 - Internet Group Management Protocol
- 2. How to efficiently distribute multicast packets from one or more sender(s) to a group of receivers ?
 - A loop-free graph must be built to connect the sender(s) to all the interested receivers
 - source tree
 - unidirectional shared tree
 - bi-directional shared tree

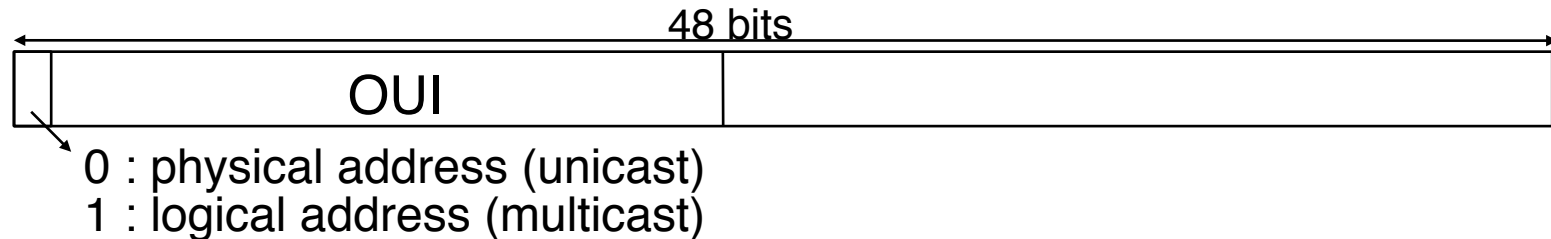
Localization of receivers



- Principle
 - Routers query their attached LANs to identify active groups
 - Hosts respond to router queries
 - Routers redistribute the localization of the receivers towards other routers

LAN-level multicast

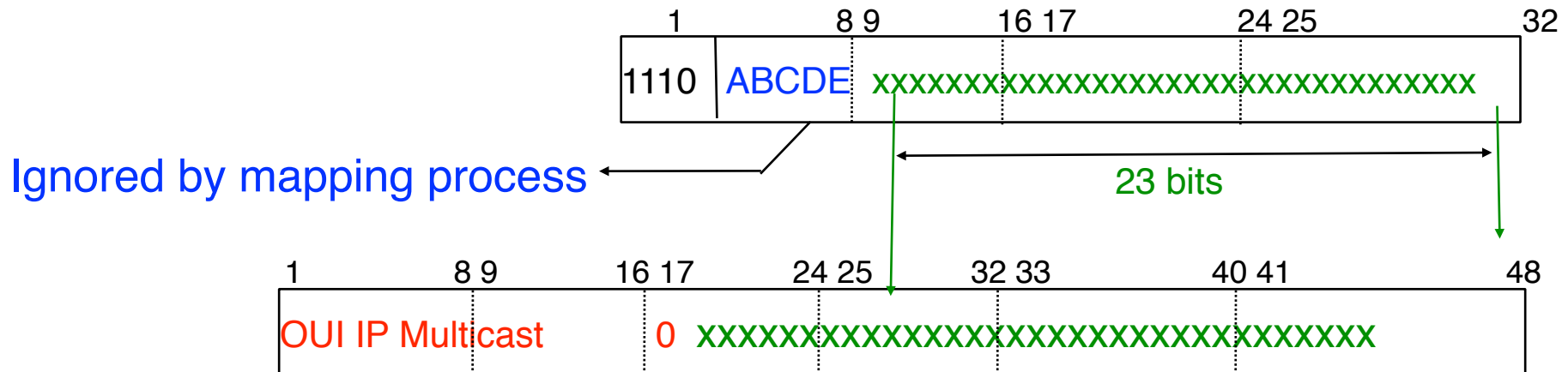
- Principle
 - Two types of destination Ethernet addresses
 - Physical addresses
 - identifies one Ethernet adapter
 - Logical addresses
 - identifies a logical group of Ethernet destinations



- Transmission of multicast frame
 - sender transmits frame with multicast destination addr.
- Reception of multicast frames
 - Ethernet adapters can be configured to capture frames whose destination address is
 - Their unicast address
 - One of a set of multicast addresses

IP Multicast over Ethernet

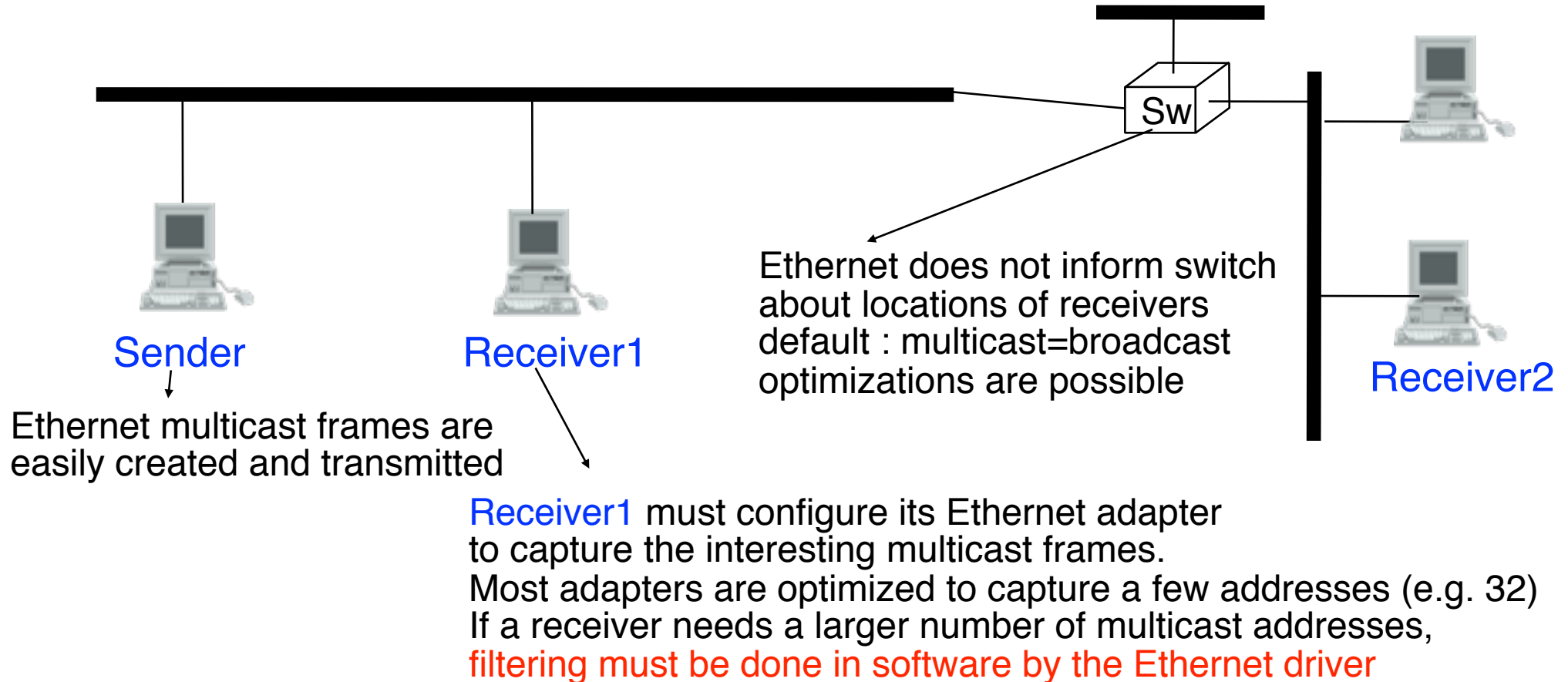
- IP Multicast over Ethernet
 - Algorithmic mapping of IP multicast addresses
 - special OUI reserved for IP multicast [0x00005e]



- The low order 23 bits of the IP multicast address are copied in the low order 23 bits of the Ethernet address
- 32 IP multicast addresses are mapped on the same Ethernet logical address

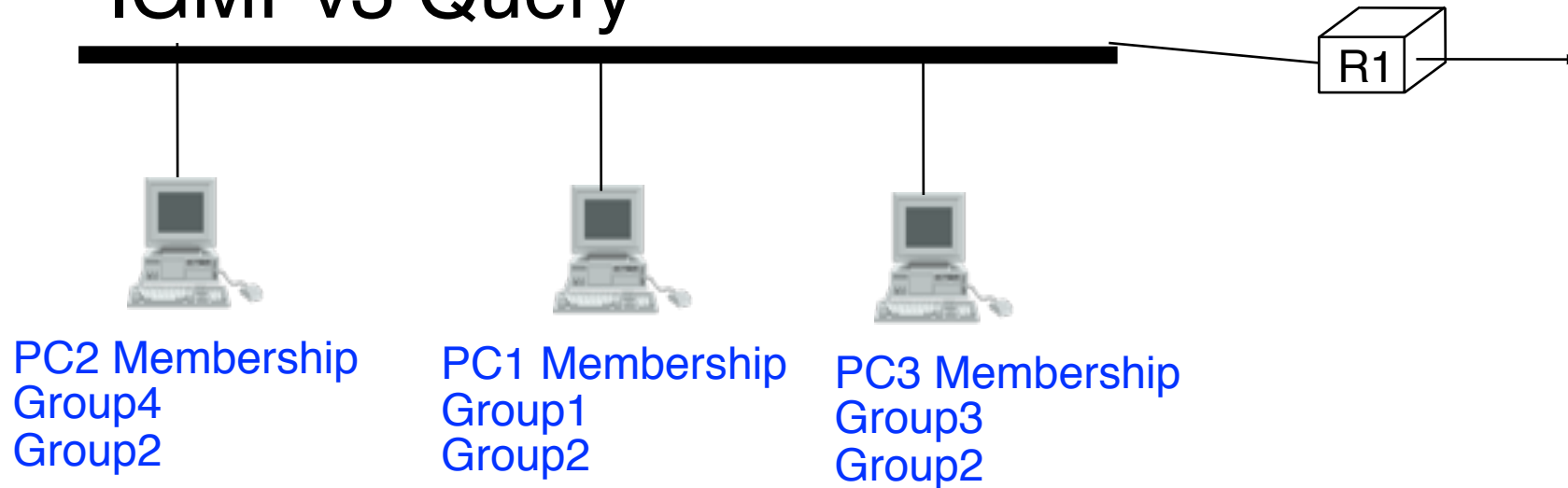
The hidden cost of LAN multicast

- Multicast on a single Ethernet
 - group of receivers identified by special address
 - receivers join group by configuring their adapter



IGMPv3

- IGMPv3 Query

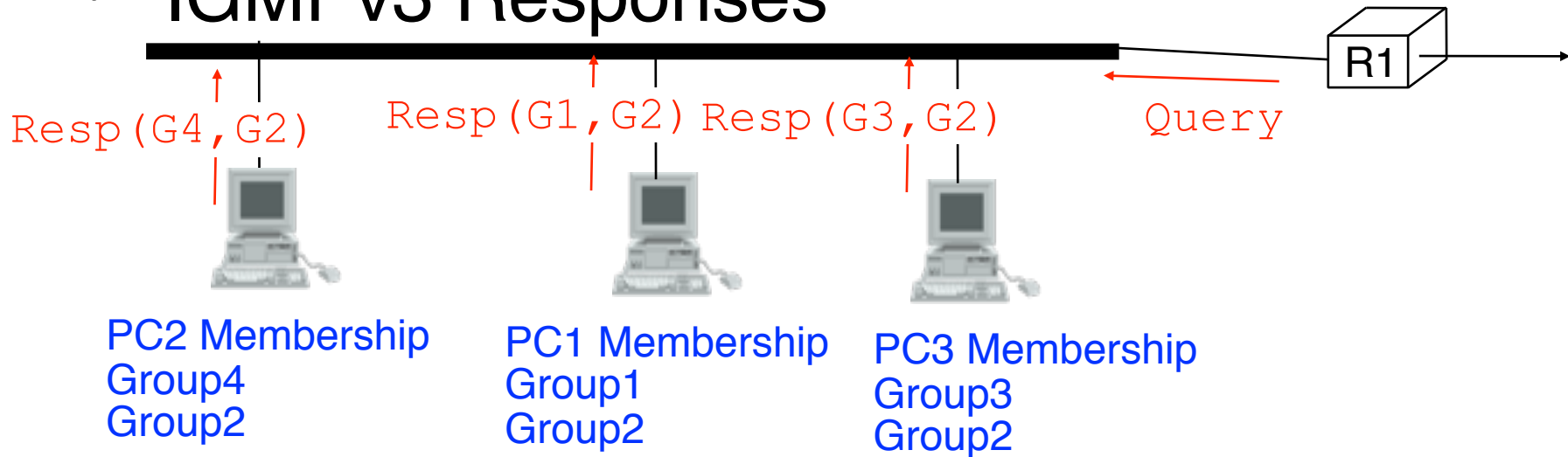


- Three types of IGM Queries

- General Query
 - similar to IGMPv1/2,
 - Group Specific Query
 - similar to Group Specific queries in IGMPv2, sent to group addr.
- Group-Source Specific Query
 - router asks responses from specific sources and groups
 - query sent to group address includes list of sources

IGMPv3 (2)

- IGMPv3 Responses

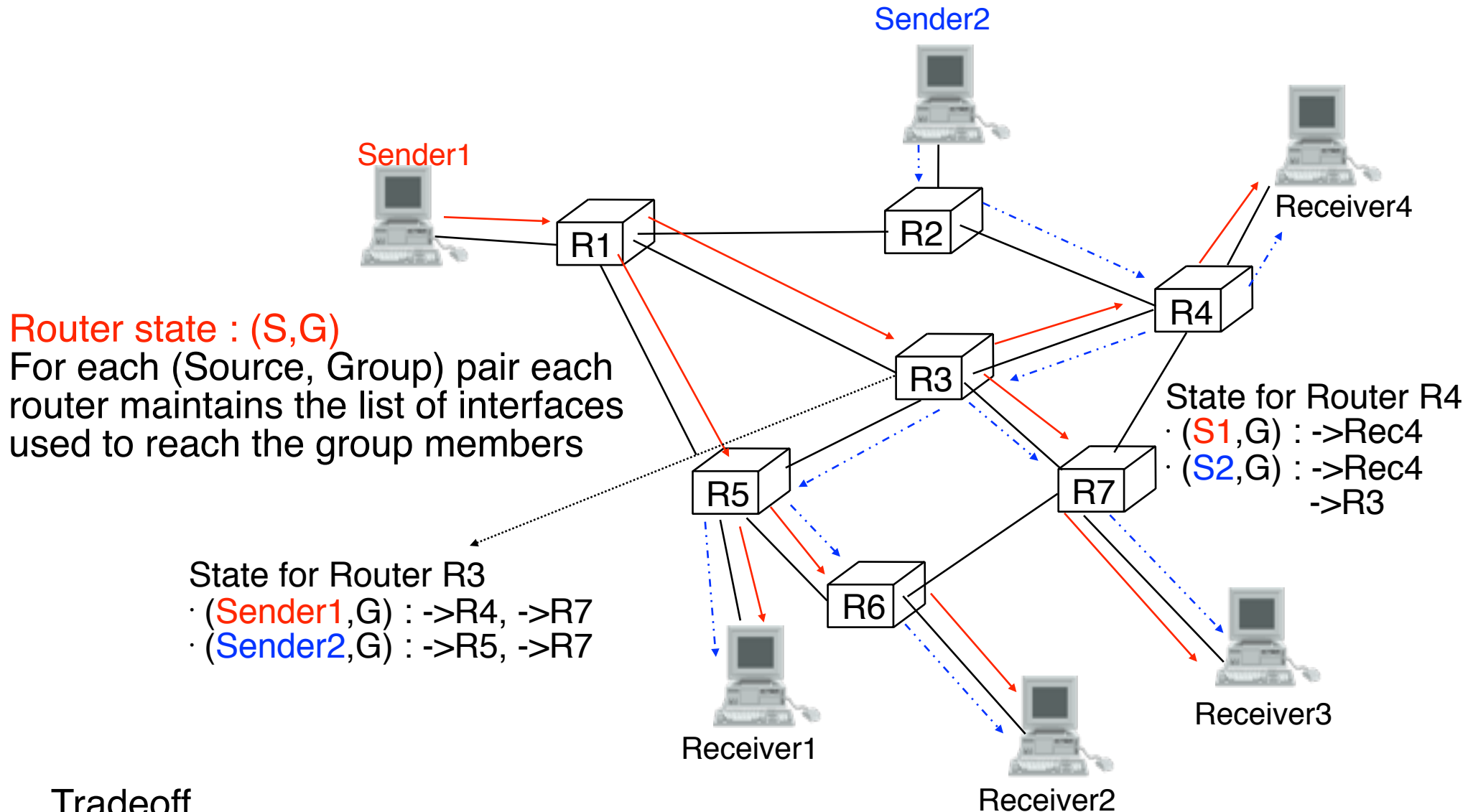


- each host sends an IGMPv3 Response
 - response contains the entire list of groups for the host
 - for each group, the host may optionally indicate that its interest is only for some specified senders (for SSM)
 - response is sent in packets with destination=224.0.0.22
 - a random timer is used before sending response to avoid overloading the network or the routers

Issues for network-level multicast

1. How to localize all the members of each multicast group ?
 - Internet Group Management Protocol
2. How to efficiently distribute multicast packets from one or more sender(s) to a group of receivers ?
 - A loop-free graph must be built to connect the sender(s) to all the interested receivers
 - • source tree
 - unidirectional shared tree
 - bi-directional shared tree

Multicast source distribution tree



Tradeoff

- optimum distribution tree versus amount of state maintained by routers

Building the source tree

- How to build the distribution tree ?

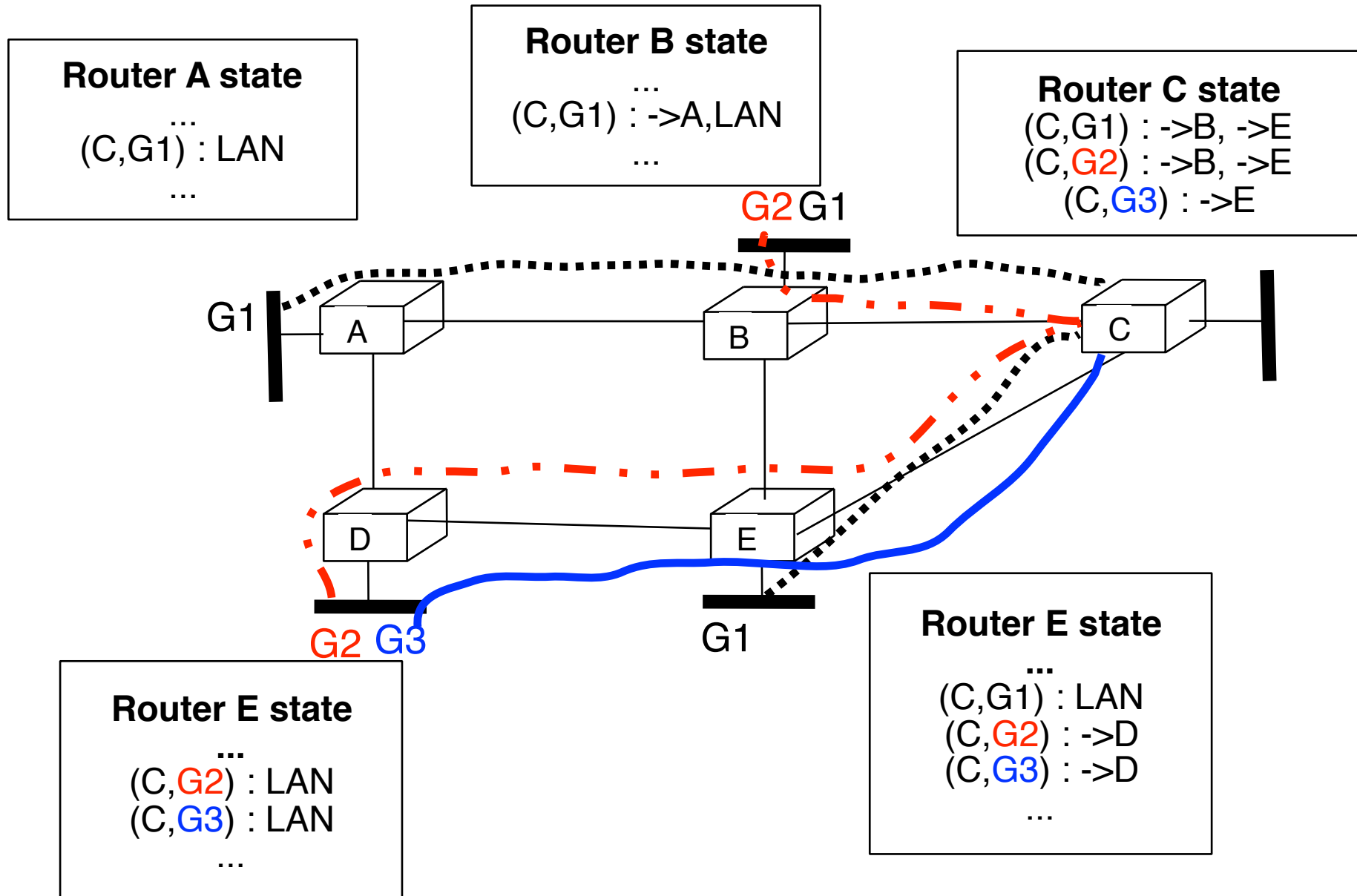
- • **Link-state solution**
 - link-state routing protocol used to distribute membership
 - Each router knows the **exact location** of group members
 - Each router can easily compute source trees
- **Broadcast solution**
 - Assume initially that group members are everywhere
 - The first packet towards a group is sent everywhere
 - If a router receives unwanted traffic, it will ask its upstream routers to stop transmitting this traffic
 - Distribution tree is built progressively

Building the source tree

Link-state solution

- Principle
 - Routers collect membership information with IGMP
 - Group membership is distributed by special type of LSPs flooded by link-state routing protocol
 - By processing those LSPs, each router knows
 - The entire network topology
 - The localization of the receivers from each multicast group
- Based on the link state database, each router computes (S,G) spanning trees for each Source, Group pair by using the Dijkstra algorithm

Multicast with OSPF



Multicast with OSPF (2)

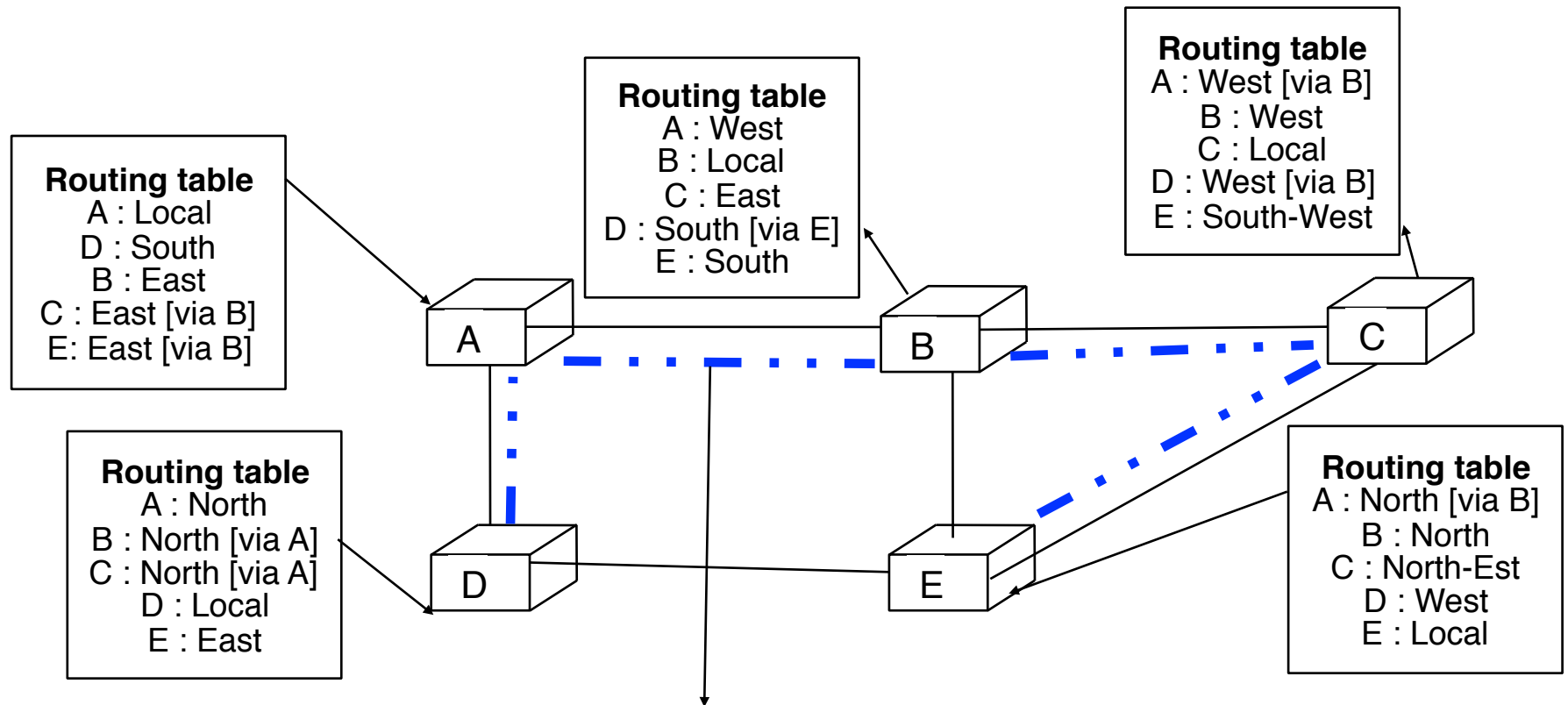
- Advantages of OSPF for IP Multicast
 - routers have complete information on
 - network topology
 - localization of group members
- Drawbacks of OSPF for IP Multicast
 - Large number of LSPs if there are many groups or many group members in the network
 - these LSPs may consume a lot of memory inside all routers whether or not there is multicast traffic
 - group membership LSPs compete with normal LSPs
 - Amount of router state depends on number of (S,G) pairs
 - Computation of (S,G) spanning trees can be costly

Building the source tree

- How to build the distribution tree ?
 - Link-state solution
 - link-state routing protocol used to distribute membership
 - Each router knows the exact location of group members
 - Each router can easily compute source trees
 - • **Broadcast solution**
 - Assume initially that group members are **everywhere**
 - The first packet towards a group is sent **everywhere**
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Broadcasting IP packets

- How to broadcast packets in a large network ?



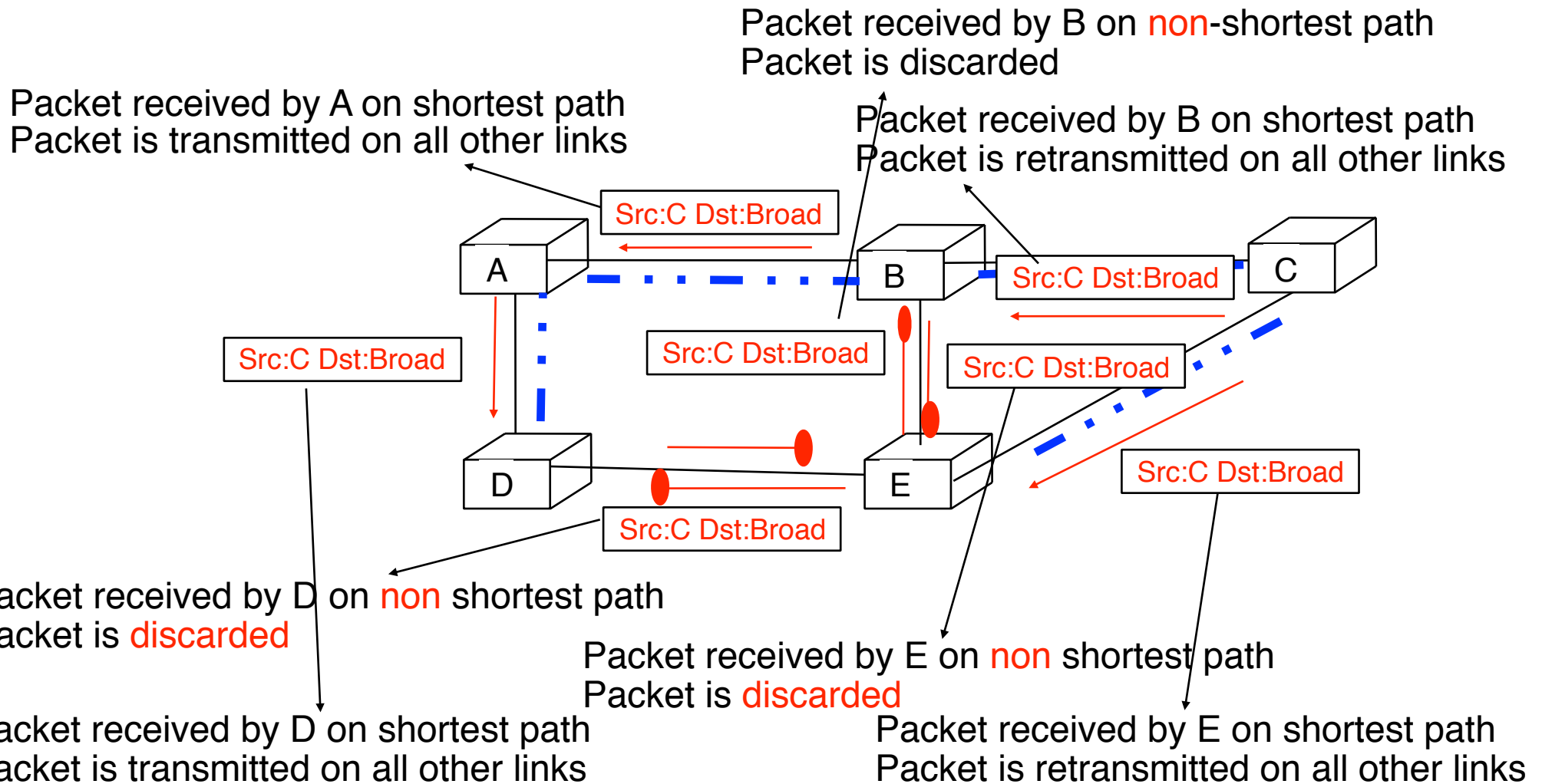
The graph containing the links from all shortest paths originating at router C creates a spanning tree rooted at C

Reverse Path Forwarding

- Principle of the solution
 - Forwarding of broadcast packets depends on
 - Multicast Destination IP address
 - Unicast Source IP address
 - Simple router forwarding algorithm

```
Broadcast packet from Source S received on interface in:
// consult routing table
// to find shortest route towards S
if (interface in == shortest_route(to S))
    for(i=0; i++; i<N)
    {
        if (i!=in) Send_packet(interface i);
    }
else
    Discard(packet);
}
```

RPF example

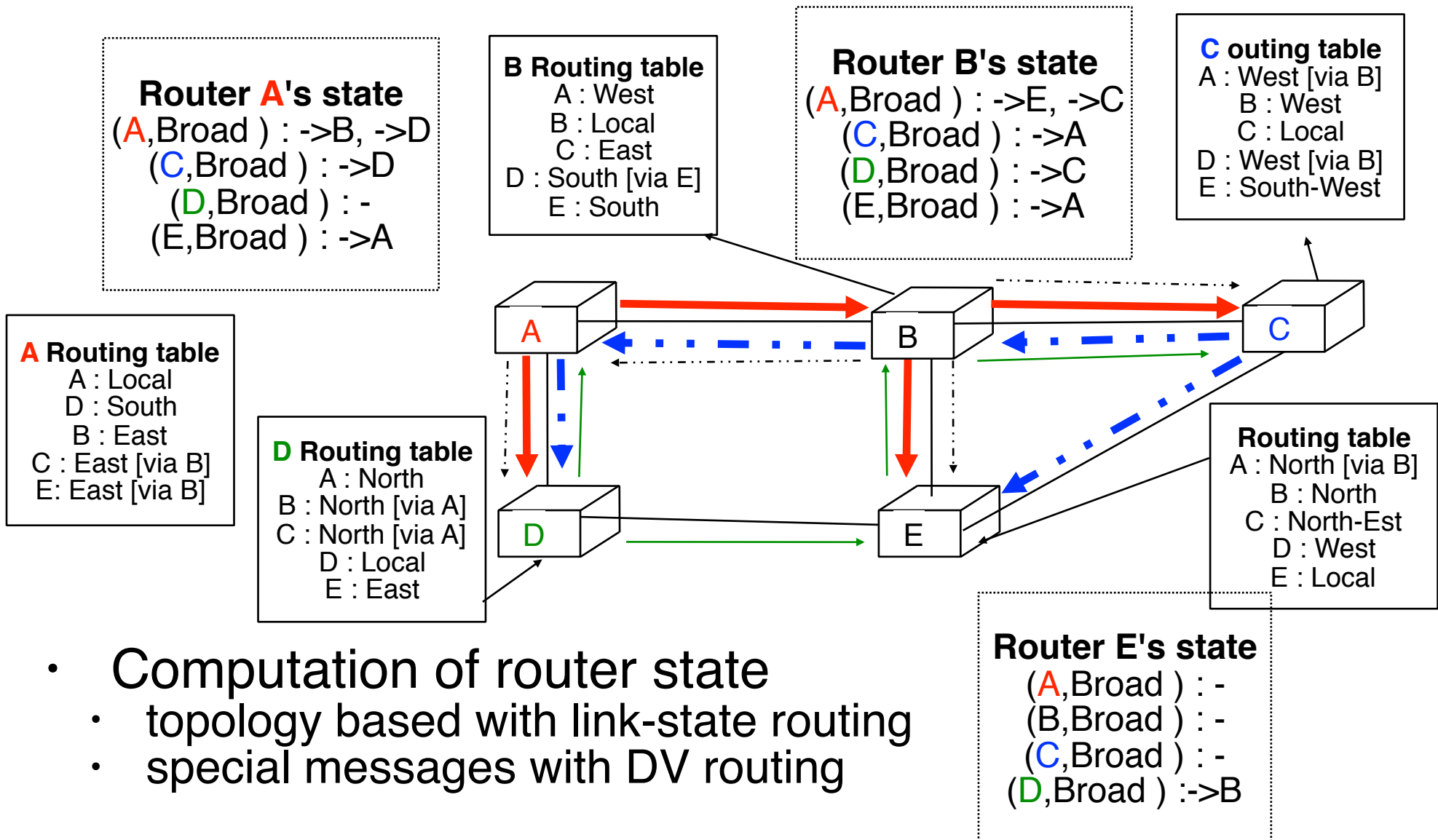


- Propagation of the broadcast packet sent by router C

Improving RPF

- How to avoid unnecessary transmissions ?
 - • **Sender-based solution**
 - for each **source**, router computes list of **outgoing interfaces**
 - broadcast packets are only forwarded on these interfaces
 - list of outgoing interfaces is computed by including interfaces towards neighbours on shortest path from source
 - router must know shortest path chosen by neighbour router
 - possible with OSPF, requires specific messages with RIP
 - CPU intensive
 - Receiver-driven solution

Improving RPF Sender-based solution



Improving RPF

- How to avoid unnecessary transmissions ?
 - Sender-based solution
 - **Receiver-driven solution**
 - • for each **source**, router maintains list of **outgoing interfaces**
 - broadcast packets are only forwarded on these interfaces
 - Computation of list of outgoing interfaces
 - Initially list of outgoing interfaces contains all interfaces
 - when a router receives a packet over a non-shortest path, it sends a special message to the upstream router
 - the upstream router will then update its list of interfaces
 - Less CPU intensive than sender-based solution

Improving RPF

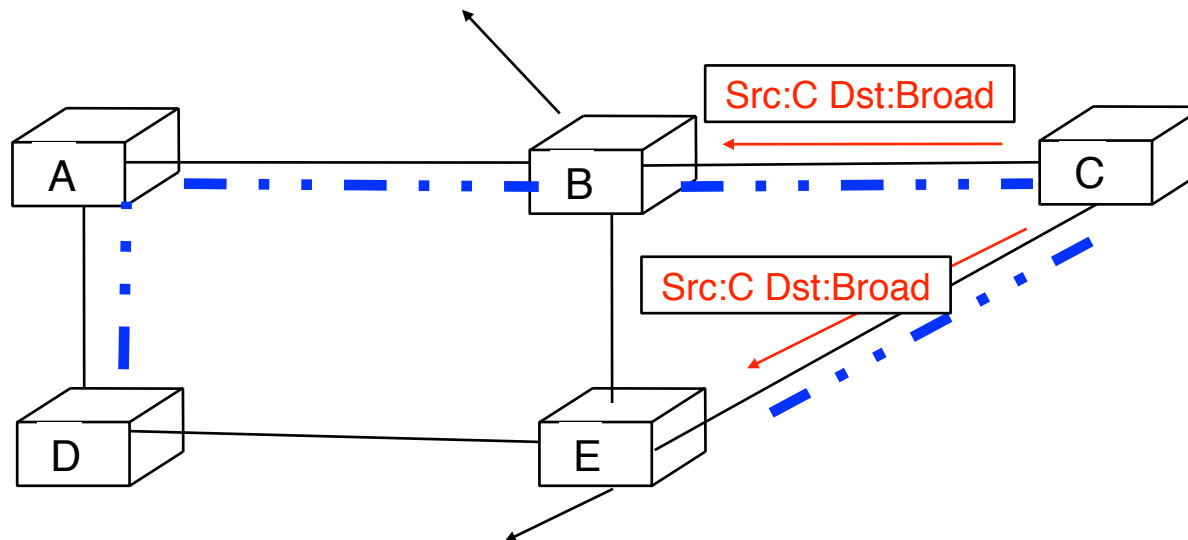
Receiver-based solution

Packet received by B on shortest path

Create state for router B:

Ⓜ (C,Broad) : ->A, ->E

Packet is retransmitted on all links in outgoing list



Packet received by E on shortest path

Create state for router E:

Ⓜ (C,Broad) : ->B, ->D

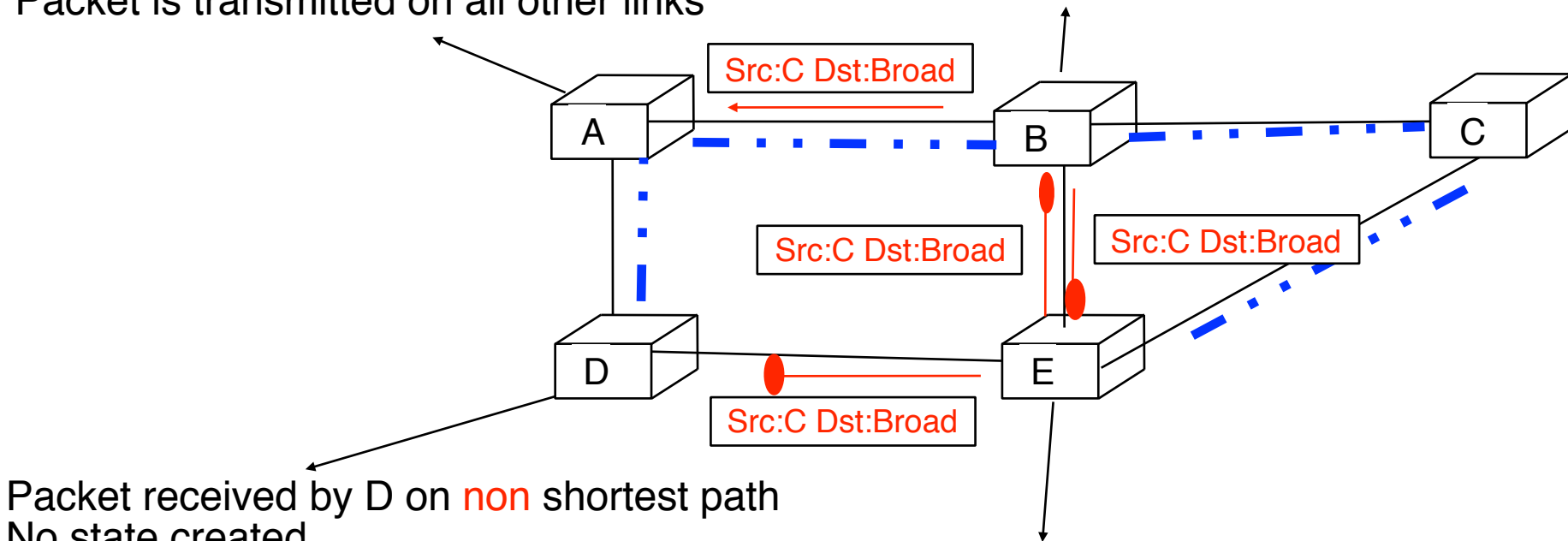
Packet is retransmitted on all links in outgoing list

Improving RPF

Receiver-based solution (2)

Packet received by A on shortest path
Create state for router A
? **(C,Broad) : ->D**
Packet is transmitted on all other links

Packet received by B from **non** shortest path
Packet is discarded
Send Prune message (C,Broad) to router E



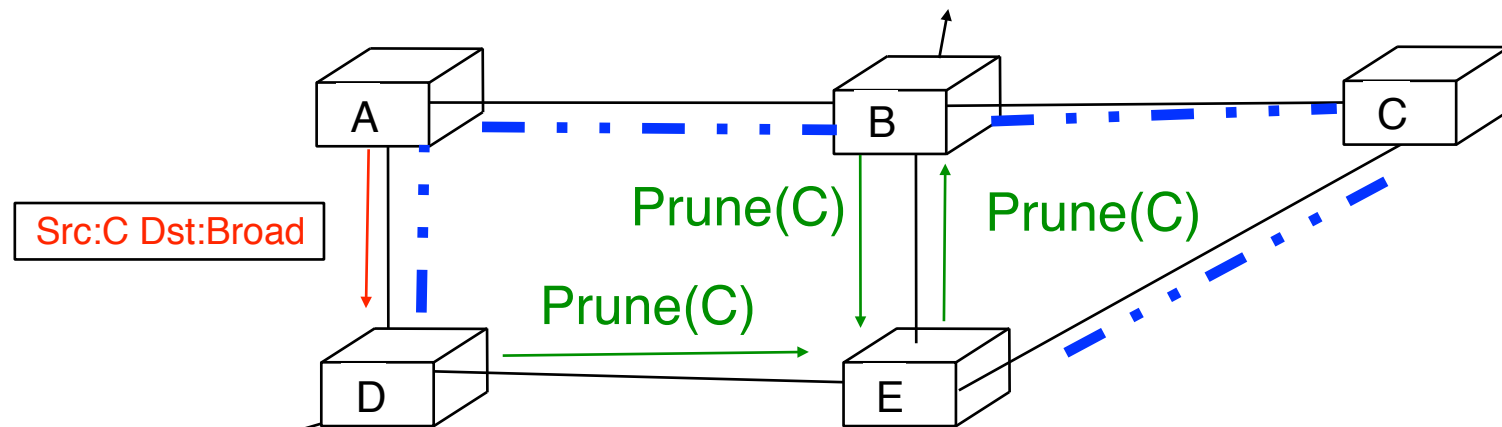
Packet received by D on **non** shortest path
No state created
Packet is **discarded**
Send Prune message (C,Broad) to router E

Packet received by E on **non** shortest path
State for router E unchanged:
? **(C,Broad) : ->B, ->D**
Packet is **discarded**
Send Prune message (C,Broad) to router B

Improving RPF

Receiver-based solution (3)

Prune message indicates that broadcast packets from C should not be retransmitted towards E
New state for router :
[?] (C,Broad) : ->A



Packet received by D on shortest path
Create state for router D :
[?] (C,Broad) : ->E
Packet is transmitted on all other links

Prune messages indicate that broadcast packets from C should not be retransmitted towards B and D
New state for router E :
[?] (C,Broad) : -

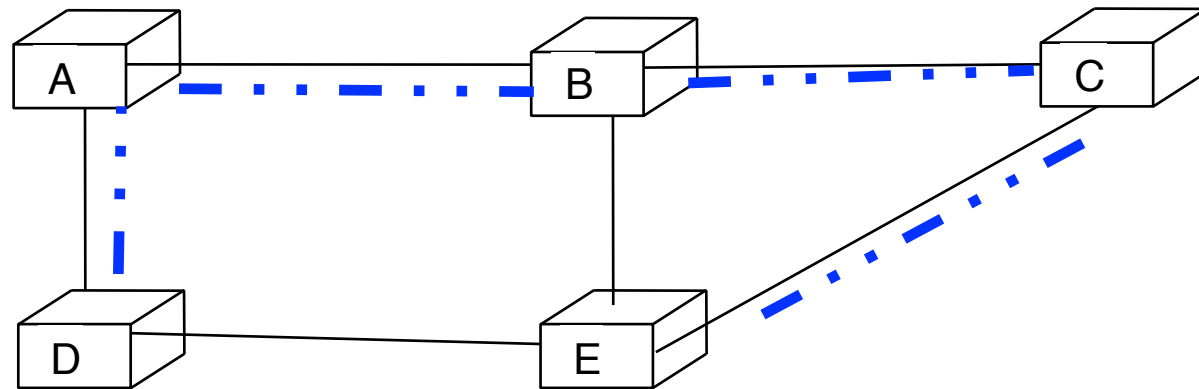
Improving RPF

Receiver-based solution (4)

- Final router states

State for router A :
[?] (C,Broad) : ->D

State for router B :
[?] (C,Broad) : ->A



State for router D :
[?] (C,Broad) : -

State for router E :
[?] (C,Broad) : -

- next broadcast packets from C will only flow on shortest path tree
- state is created dynamically when multicast packets are received on shortest path from source
- a timer is associated with each state
 - state is removed when timer expires

Using RPF for Multicast

- Principles
 - Multicast packets are initially transmitted by routers as if they were broadcast packets
 - Routers create (S,G) state upon reception of the first multicast packet from S to G
 - Two types of Prune messages are sent
 - Source specific Prune (S,G) message if packet is received on non-shortest path
 - Group Prune message if there is no group member downstream of this router
 - (S,G) state is updated by reception of Prune msgs

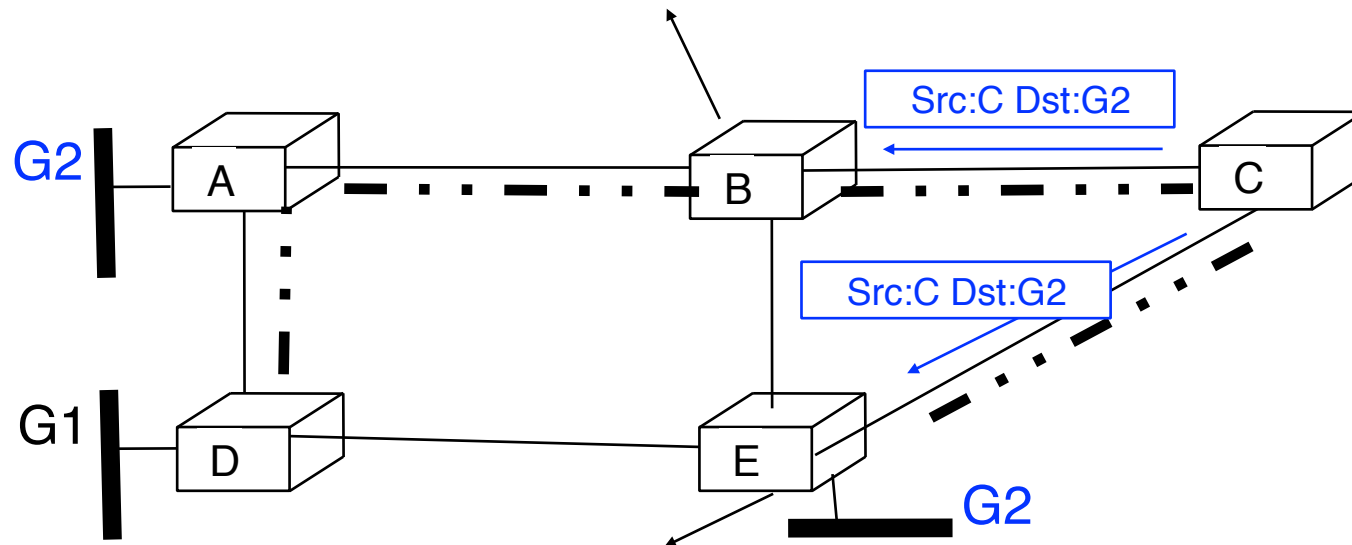
RPF for Multicast : Example

Packet received by B on shortest path
(C, **G2**) state is set to all router interfaces except incoming interface

New state for router B:

⊡ (C, **G2**) : ->A, ->E

Packet is retransmitted on all branches of source tree



Packet received by E on shortest path

(C, **G2**) state is set to all router interfaces except incoming interface

New state for router B:

⊡ (C, **G2**) : ->B, ->D, LAN

Packet is retransmitted on all branches of source tree

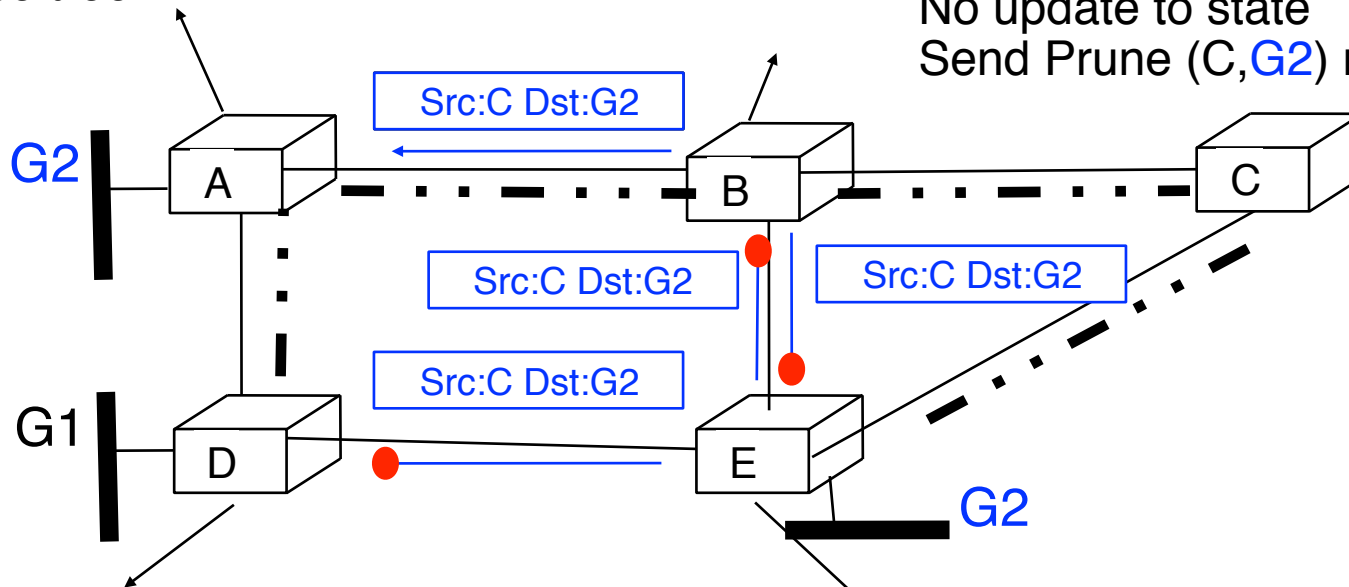
RPF for Multicast : Example (2)

Packet received by A on shortest path
(C,G2) is set to all interfaces except incoming interface
New state for router A:

■ (C,G2) : ->D, LAN

Packet is retransmitted on all branches
of source tree

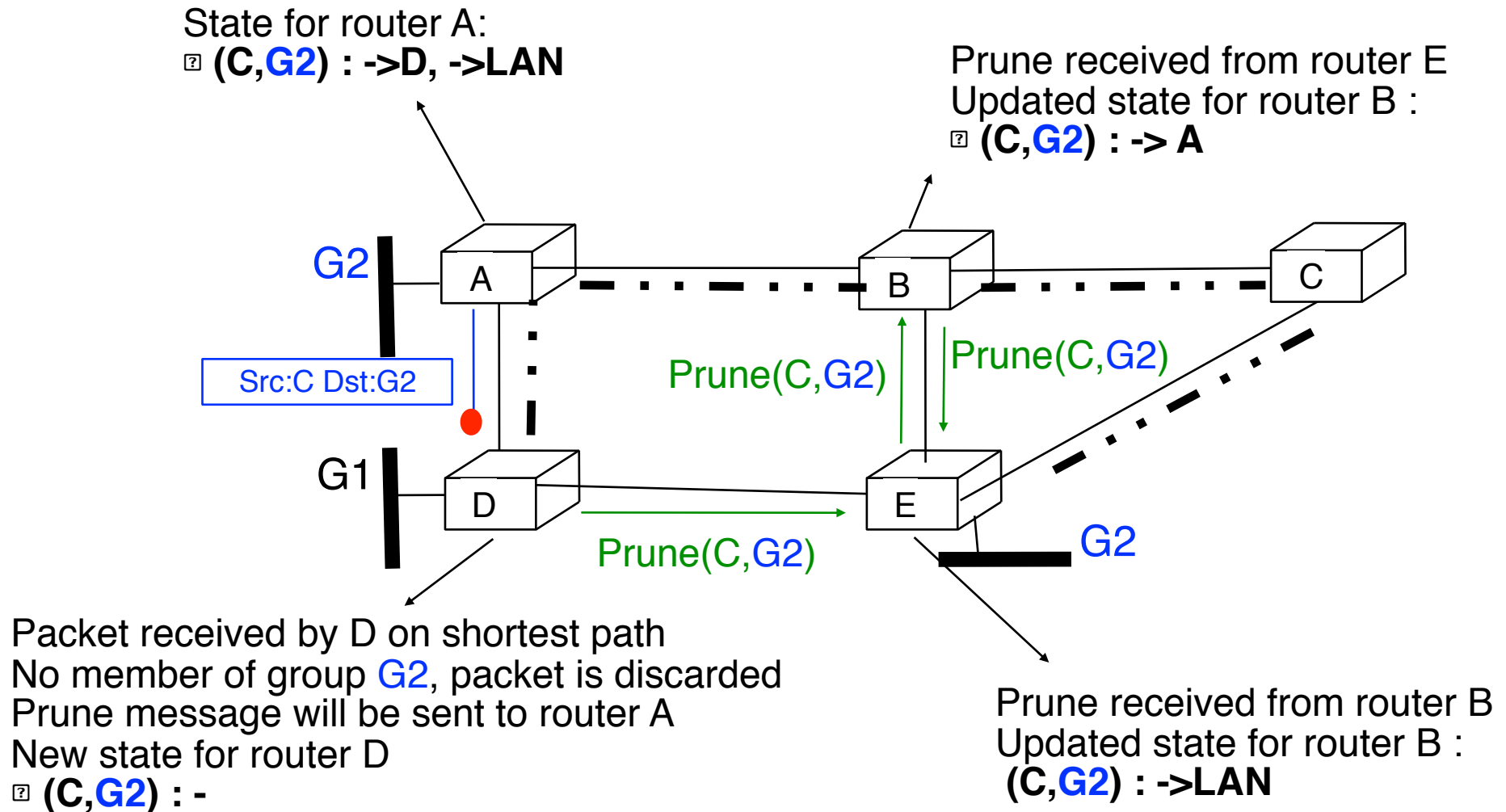
Packet received by B on **non** shortest path
Packet is discarded
No update to state
Send Prune (C,G2) msg to E



Packet received by D on **non** shortest path
Packet is discarded
No state is created in router D
Send Prune (C,G2) msg to E

Packet received by E on **non** shortest path
Packet is discarded
No update to state
Send Prune (C,G2) to B

RPF for Multicast : Example (3)



RPF for Multicast : Example (4)

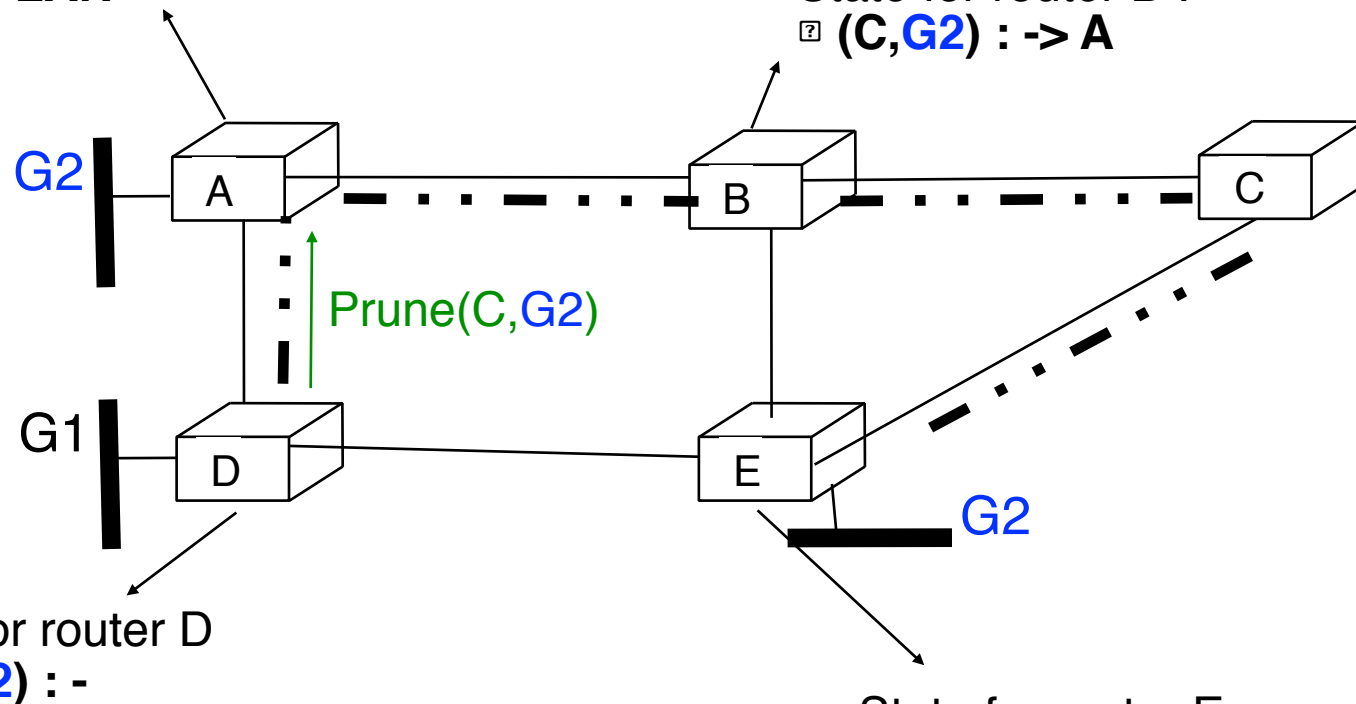
Prune received from router B

New state for router A :

(C,G2) : ->LAN

State for router B :

(C,G2) : -> A



State for router D

(C,G2) : -

State for router E :

(C,G2) : LAN

[?] State lifetime

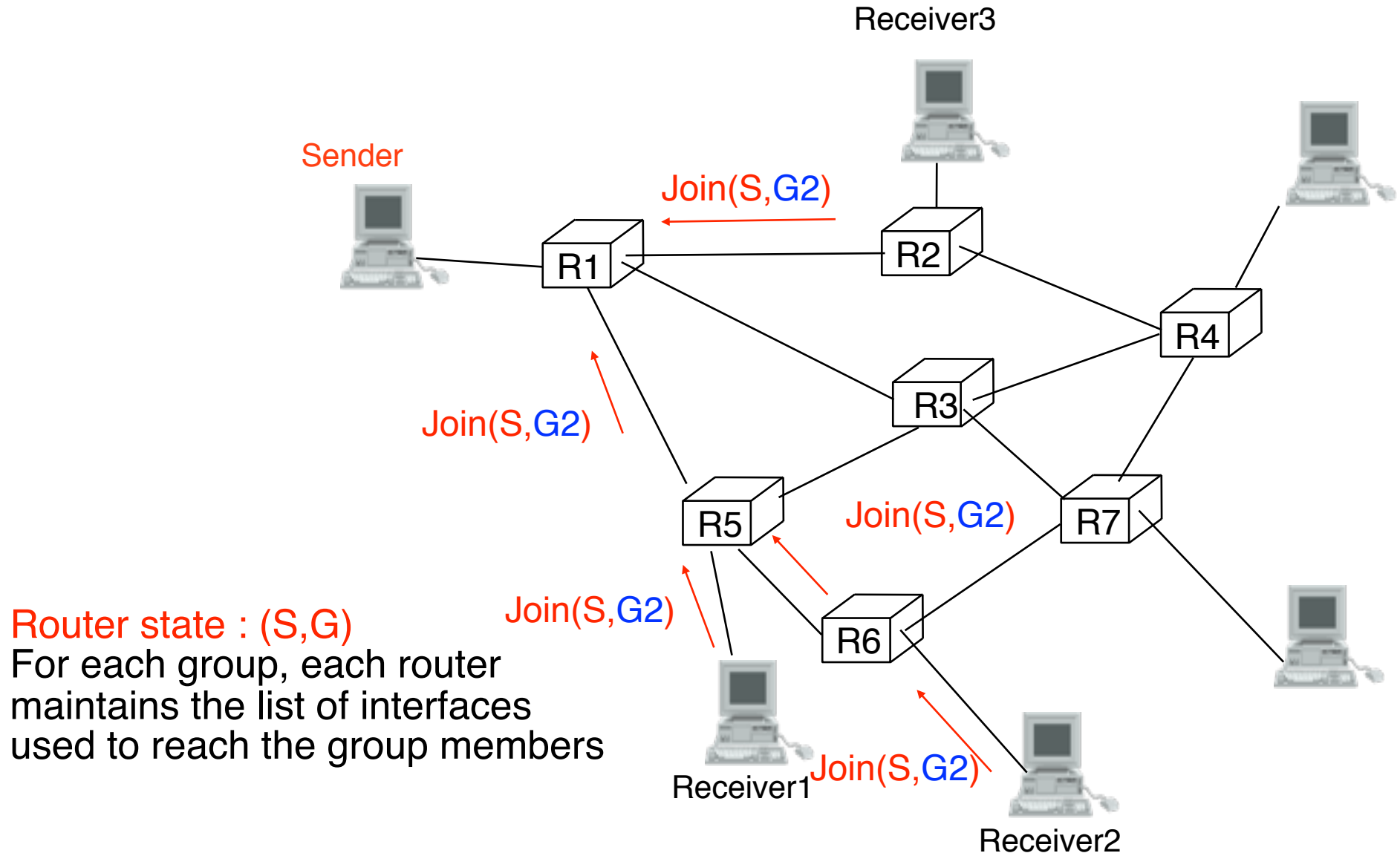
- [?] periodically state info is removed and flooding restarts
- [?] flood and prune operation

[?] Graft msg can be used by D to join the tree later

Example protocols

- DVMRP
 - first multicast routing protocol deployed
 - special unicast distance vector routing protocol
 - sender-based optimisation for RPF
 - special poison reverse message to inform upstream router that it is the best path for a given source
 - timer associated with router state
 - periodic flood and prune
- PIM-Dense Mode
 - periodic flood and prune
 - does not rely on a special unicast routing protocol
 - RPF is based on existing unicast routing table
 - receiver-based optimisation for RPF

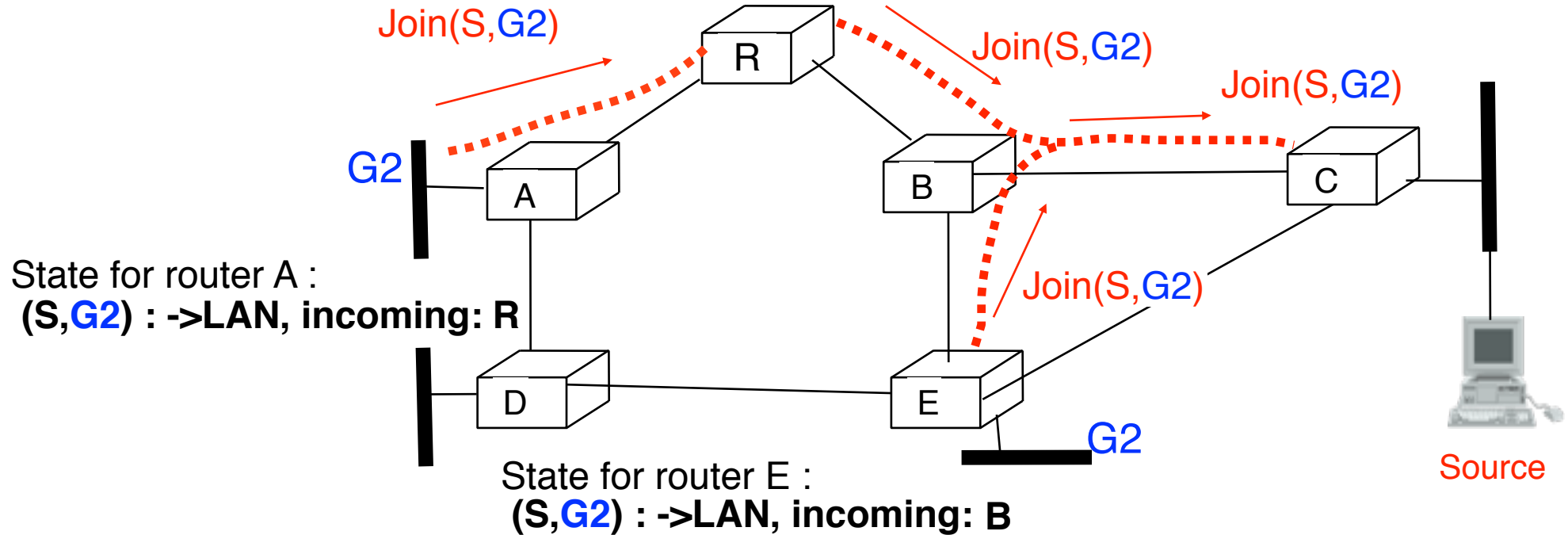
Source tree for SSM



Building the source tree

State for R:
(S, **G2**) : ->A, incoming : B

State for router B :
(S, **G2**): ->R ->E, incoming : C



Issues for network-level multicast

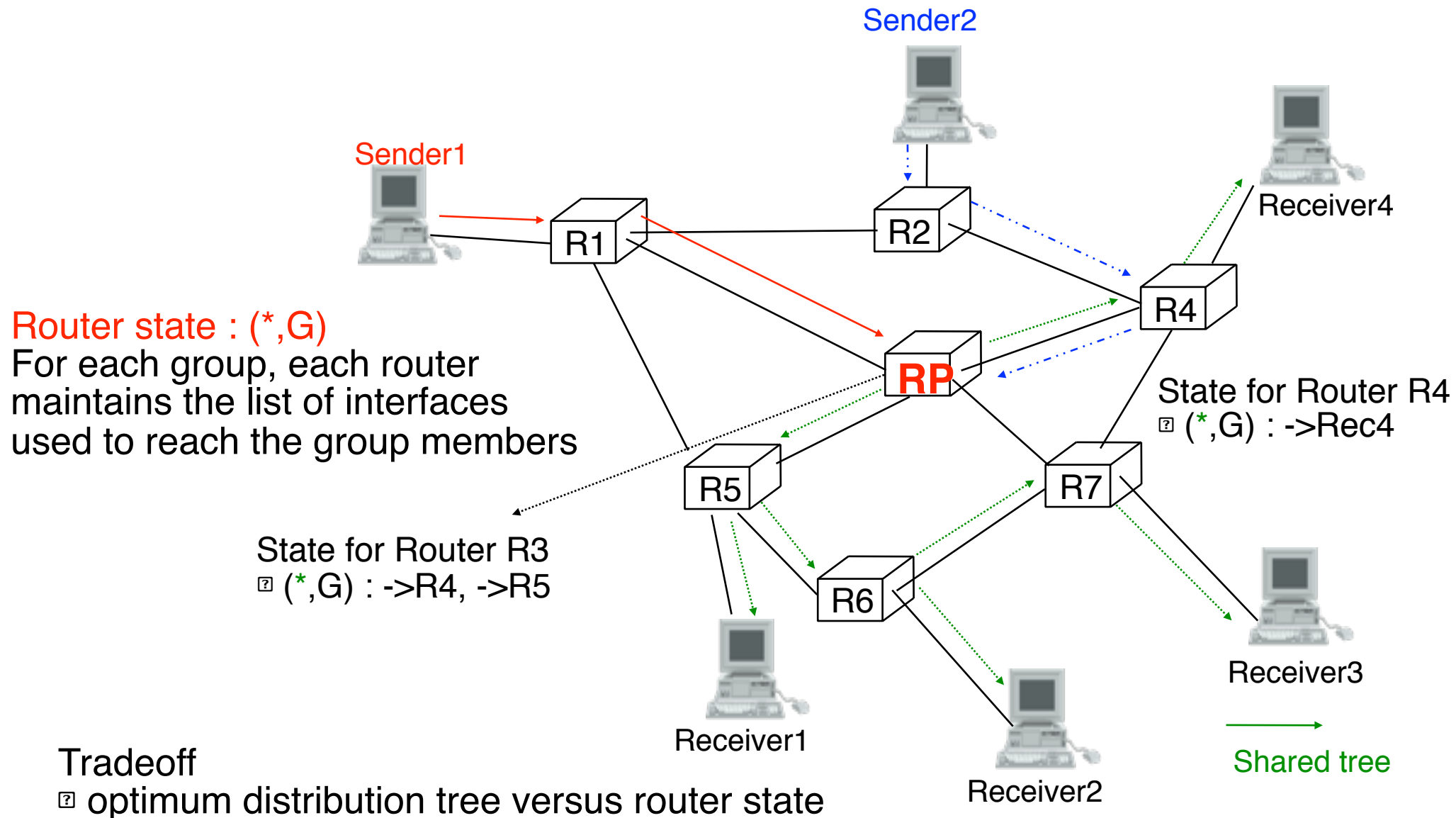
1. How to localise all the members of each multicast group ?

☐ Internet Group Management Protocol

2. How to efficiently distribute multicast packets from one or more sender(s) to a group of receivers ?

- A loop-free graph must be built to connect the sender(s) to all the interested receivers
 - source tree
 -  unidirectional shared tree
 - bi-directional shared tree

Multicast shared distribution tree



Using unidirectional shared trees

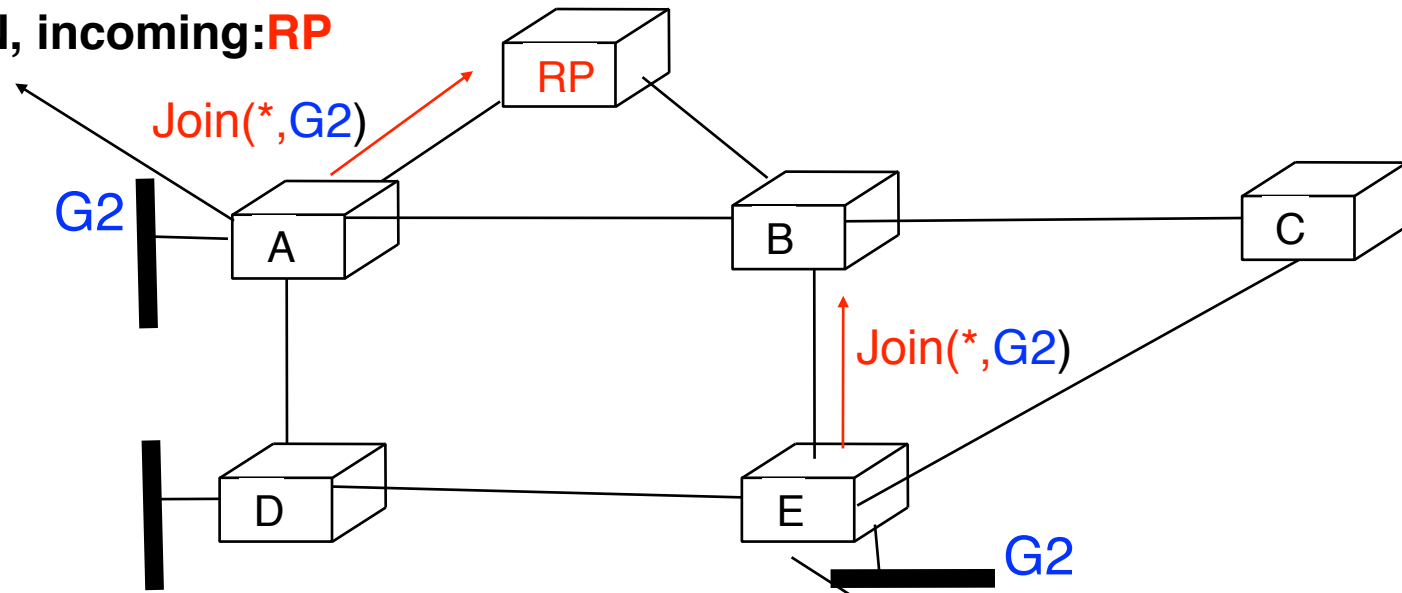
- Principles
 - One router is configured as **R**endez-vous **P**oint
 - All other routers know the address of the **RP**
 - **RP** is the root of the shared $(*,G)$ tree
 - Multicast packets sent by hosts are intercepted by multicast routers and sent to the **RP**
 - **RP** redistributes these packets over the $(*,G)$ tree
 - Receivers join dynamically the shared $(*,G)$ tree

Construction of the shared tree

A has a member of group **G2**

Initial state for router A :

(*,G2) : ->LAN, incoming:RP



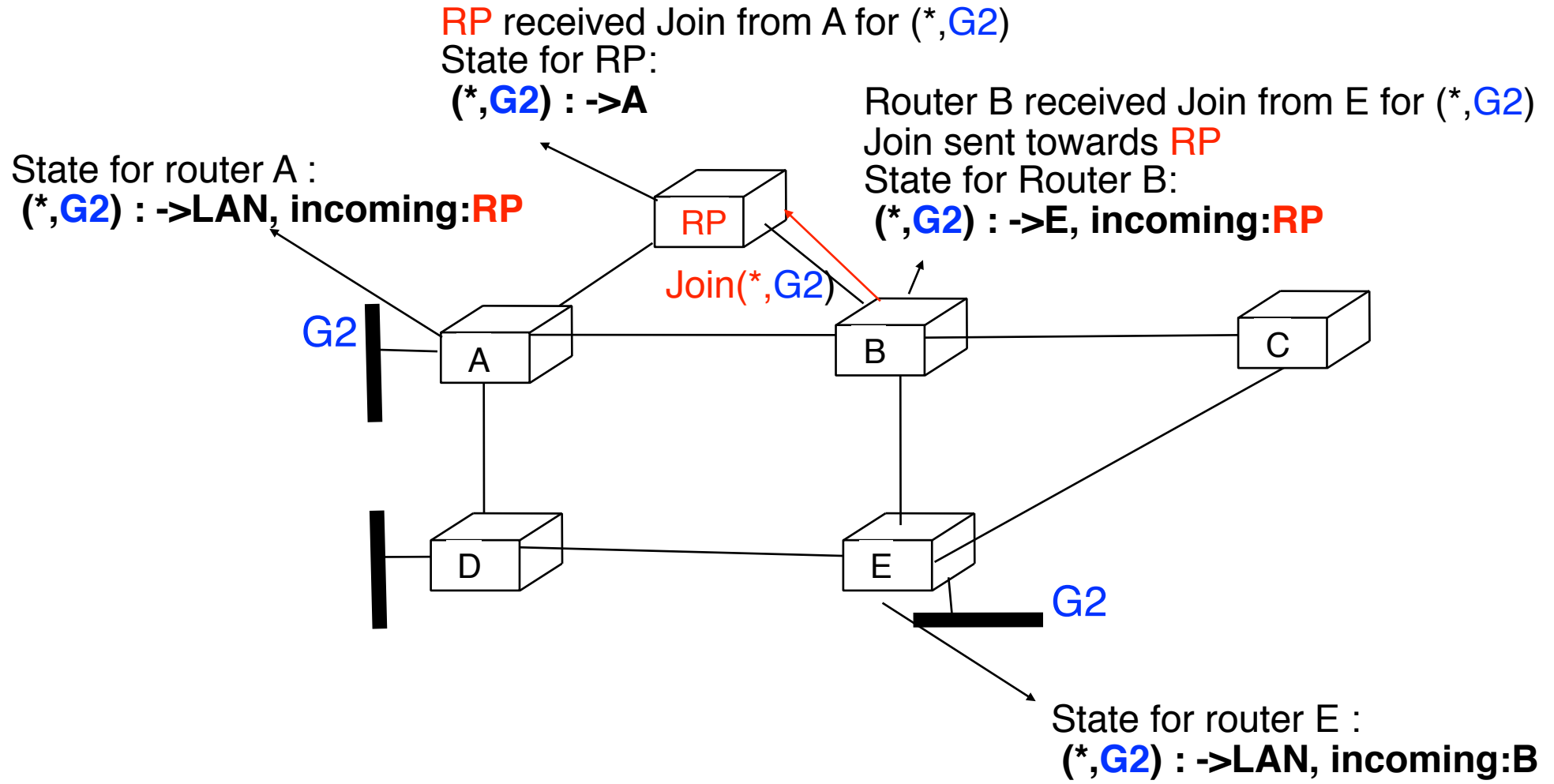
E has a member of group **G2**

Initial state for router E :

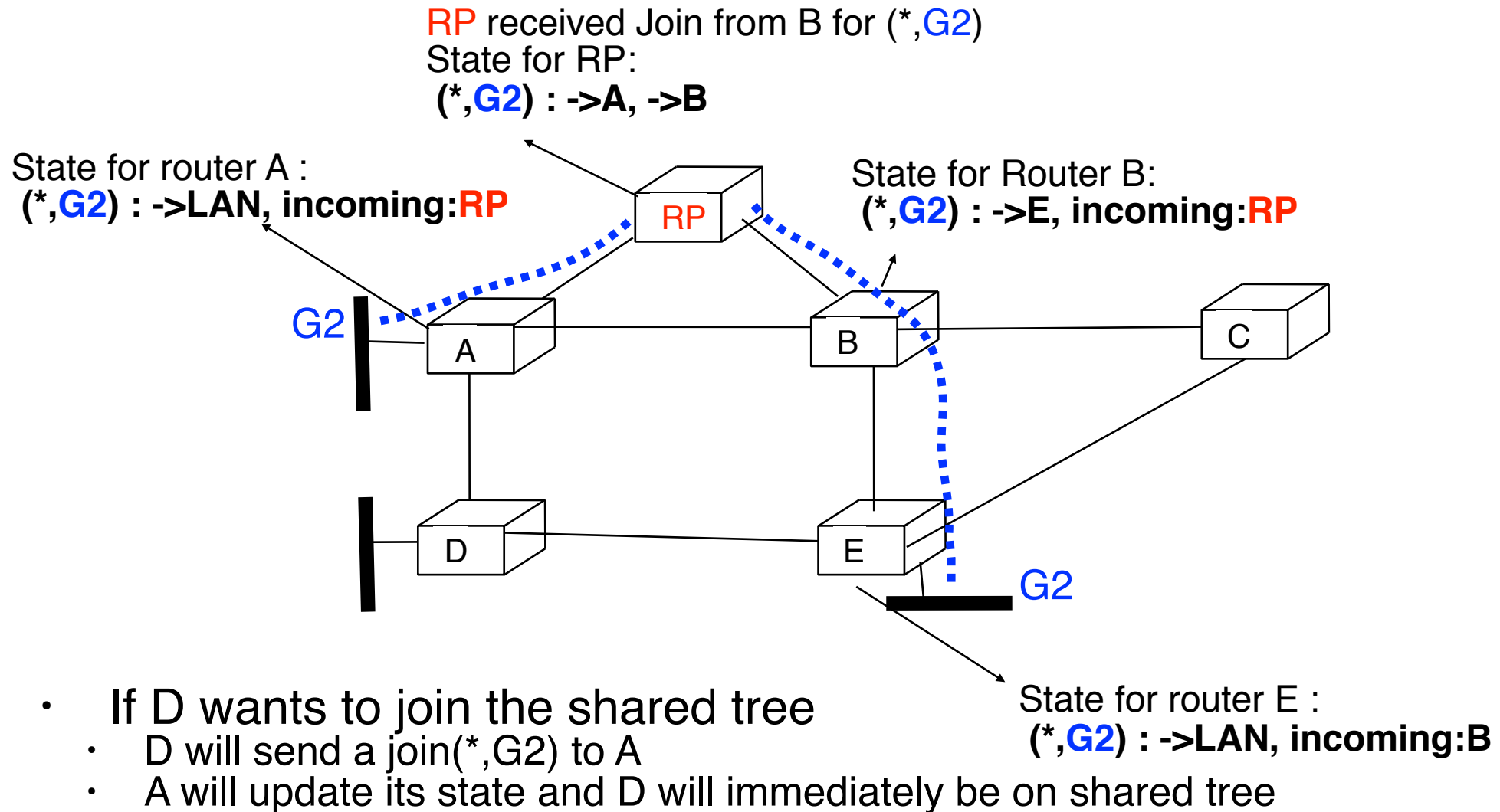
(*,G2) : ->LAN, incoming:B

- Routers attached to group members send Join (*,G) msgs towards RP to build shared tree

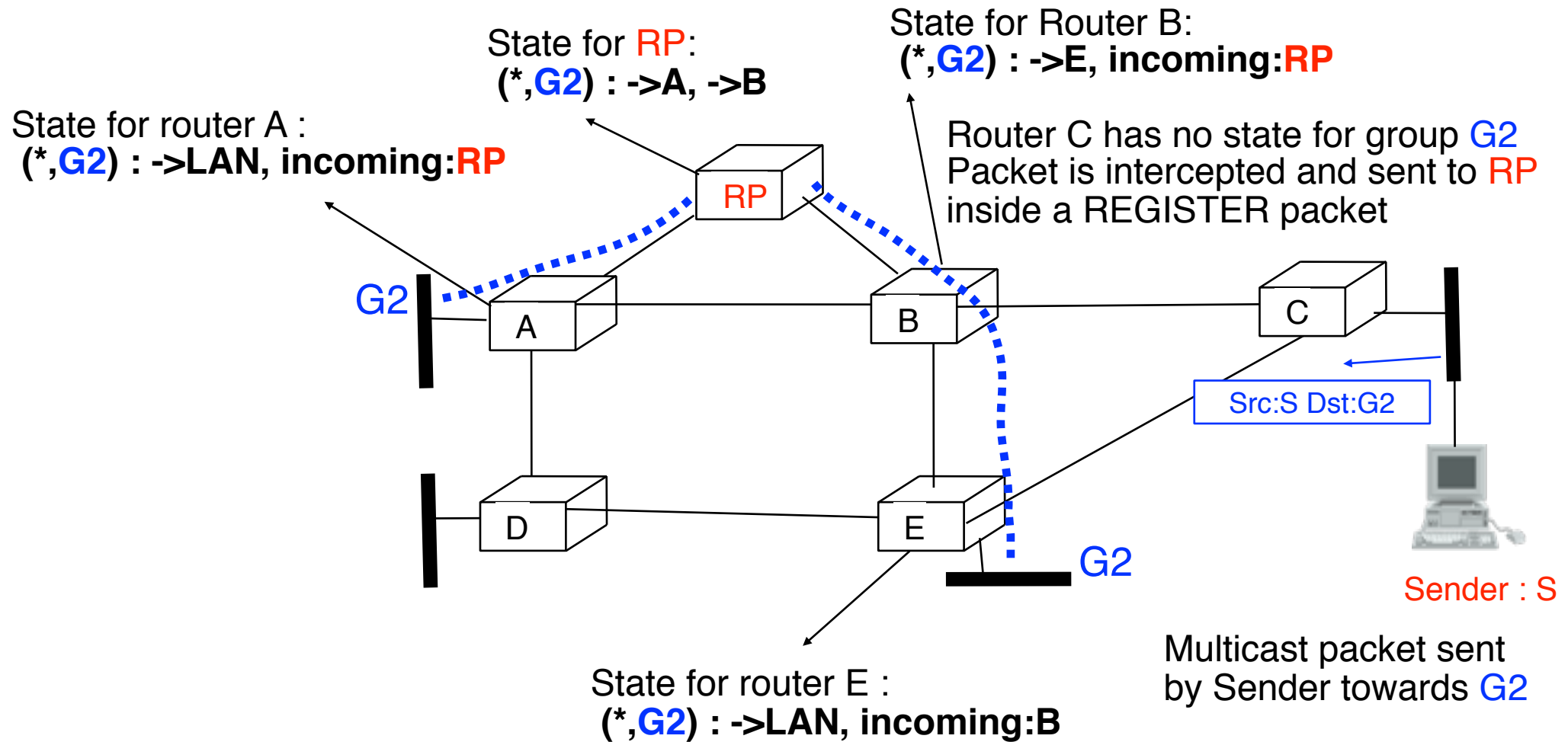
Construction of the shared tree (2)



Construction of the shared tree (3)

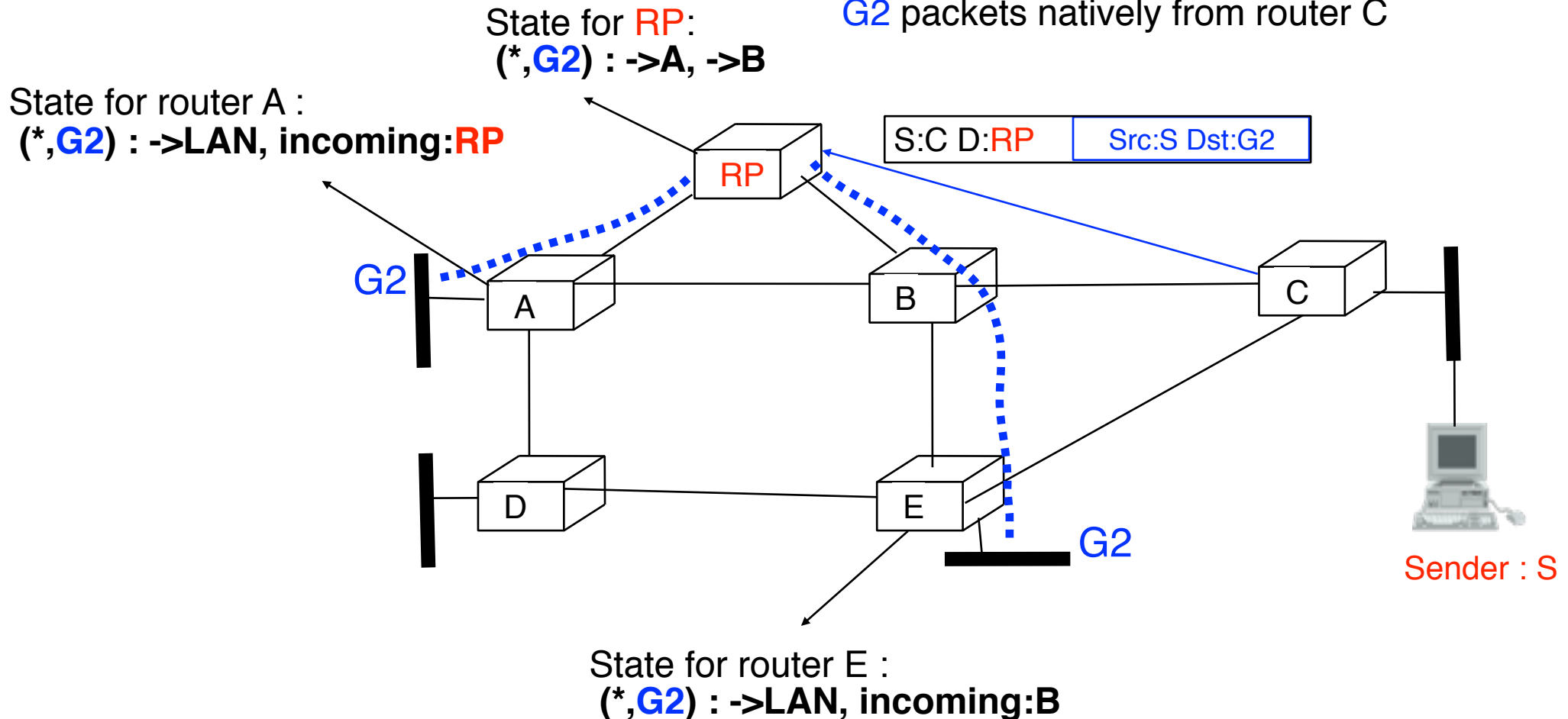


Transmission of multicast packets

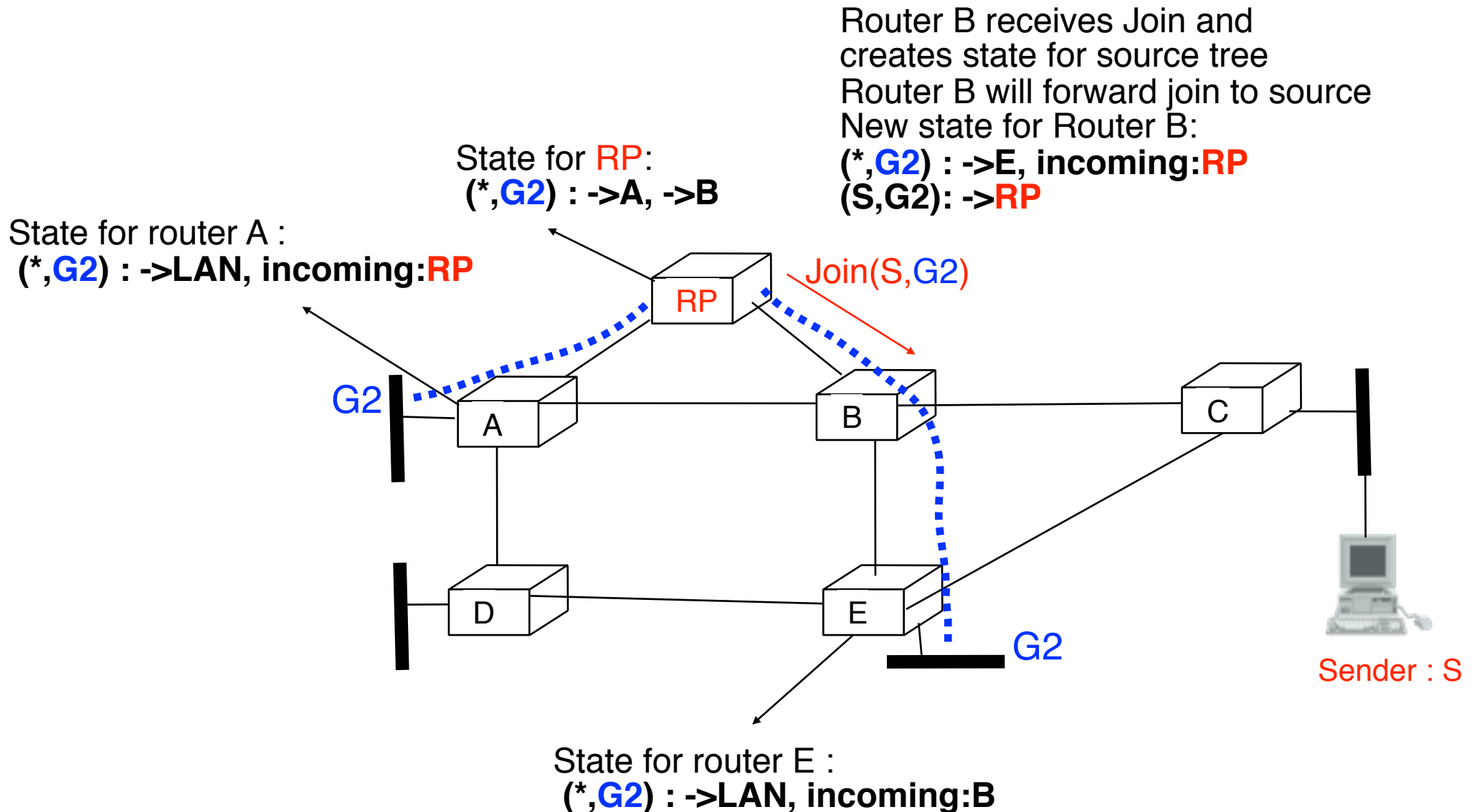


Transmission of multicast packets (2)

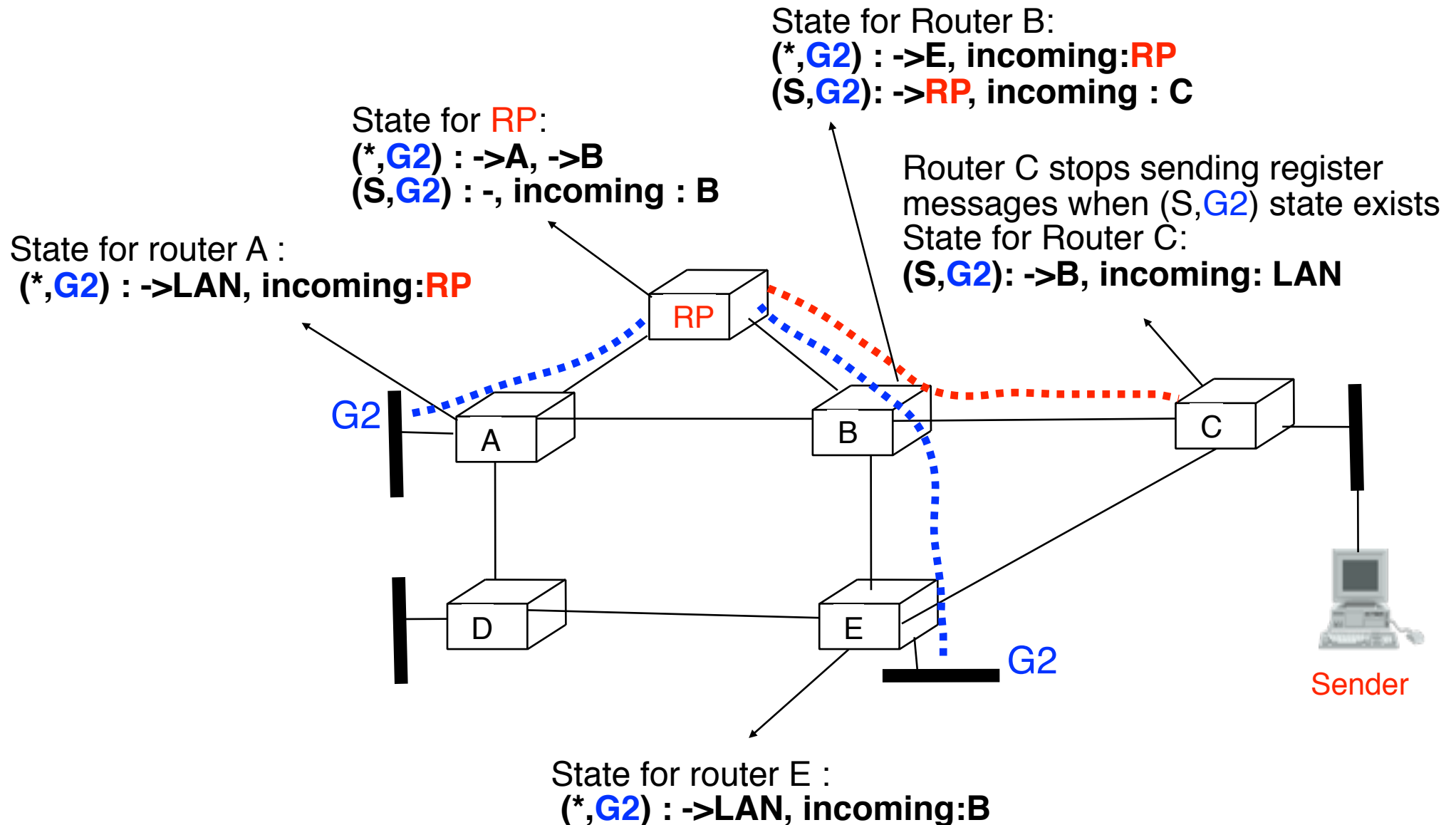
RP checks that there is state for G2
Packet is de-encapsulated and
forwarded on (*,G2) shared tree
RP will create a source tree to receive
G2 packets natively from router C



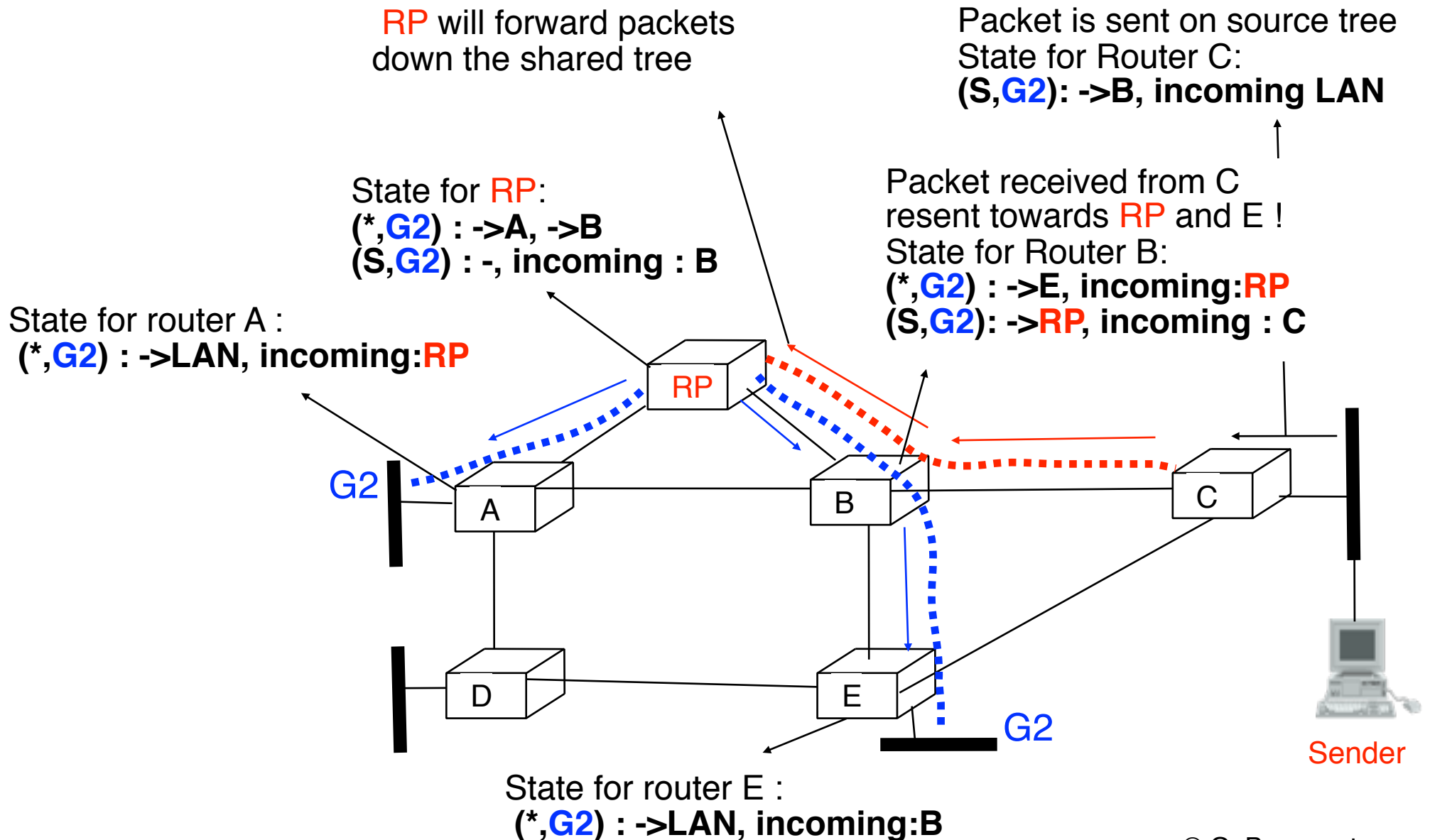
Construction of the source tree



Construction of the source tree (2)



Forwarding of multicast packets



PIM Sparse Mode

- Widely deployed multicast routing protocol
- Possible configurations of Rendez-vous Points
 - RP addresses manually configured
 - using anycast to discover Rendez-vous Point
 - RP broadcast announces in dense-mode
- Possible utilisation of unidirectional shared tree
 - For all multicast packets
 - But RP could become overloaded with multicast
 - For first multicast packets sent to each group
 - Those packets are first sent to RP to let receivers hear from sources
 - first-hop routers can join the source-tree if there is a lot of traffic from a given source
 - popular PIM-SM implementation switches to source-tree as soon as traffic is >0

Using bi-directional shared trees

- Principles
 - One router is configured as **core** for each group
 - All other routers know the address of the **core**
 - **Core** is root of bi-directional shared tree
 - Bi-directional shared tree is used to forward multicast packets sent by any source
 - Routers join dynamically the shared tree

Multicast shared distribution tree

